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REVIEW

# Biomass-derived activated carbon/chitosan hydrogel composites for adsorption of organic and inorganic pollutants

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## Abstract

Water pollution caused by both organic and inorganic contaminants has emerged as a critical global concern, promoting the need for effective and environmentally sustainable treatment solutions. Among the promising approaches, biomass derived activated carbon and chitosan hydrogels have attracted significant attention due to their renewable origins, strong pollutant adsorption abilities, and adaptable properties. Biomass-derived activated carbon is particularly valued for its exceptionally large surface area and high affinity towards wide range of contaminants making it a cost effective and sustainable choice for water purification. Meanwhile, chitosan hydrogels stand out for their natural biocompatibility and can be easily modified to suit specific treatment needs. This review highlights recent advances in developing composite materials that merge activated carbon with chitosan hydrogels, focusing on synthesis routes, structural features, and their performance in wastewater remediation. It examines fabrication methods designed to improve mechanical strength and adsorption efficiency of these composites, along with operational factors such as pH levels, temperature, initial pollutant concentration, dosage of adsorbent, and ionic strength that influence adsorption behavior. The combination of activated carbon with chitosan hydrogels demonstrates synergistic benefits, including superior adsorption capacity and faster pollutant removal rates. The review also discusses ongoing challenges such as material durability, ensuring regeneration, and outlines future research directions to support the broader application and scalability of these materials. By consolidating current findings and identifying gaps in knowledge, this review provides a detailed and insightful perspective on the potential of biomass-based activated carbon and chitosan hydrogels as sustainable tools for environmental remediation.

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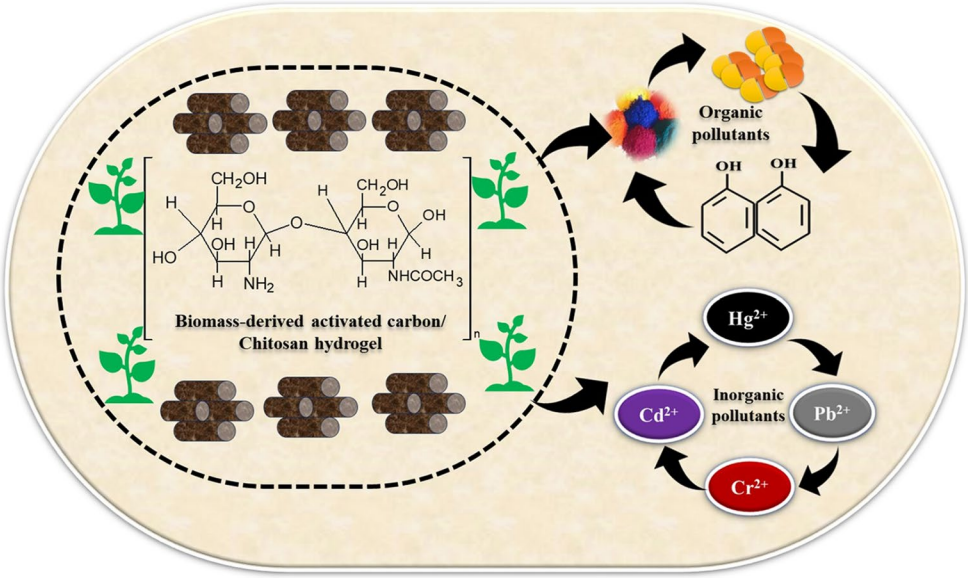
**Graphical abstract**




## Highlights

- Biomass-AC/chitosan hydrogel composites developed
- Advanced methods for composite fabrication
- Key factors optimizing adsorption efficiency
- Literature gaps and research challenges discussed
- Future directions for sustainable adsorbents

**Keywords** Activated carbon · Hydrogel · Contaminants · Chitosan · Adsorption characteristics

**Abbreviations**

| | |
|---|---|
| AC | Activated carbon |
| ACCB | Activated carbon chitosan beads |
| AO | Acid orange |
| CH | Chitosan hydrogel |
| CS | Chitosan |
| $CO_2$ | Carbon dioxide |
| $C_{CH\text{-}TS}$ | Activated carbon derived from chitosan hydrogel and tea saponin |
| DB78 | Direct blue78 |
| DIF | Diclofenac |
| HPC | Hierarchically porous carbon |
| KOH | Potassium hydroxide |
| MB | Methylene blue |
| PNP | Para-nitrophenol |
| PPCPs | Pharmaceuticals/Personal care products |
| SL | Sodium lignosulfonate |
| TC | Tetracycline |
| XPS | X-ray photoelectron spectroscopy |

## Introduction

Recently, the growing environmental threats to human survival have become increasingly alarming, largely due to the contamination of both water and the atmosphere. A notable instance of harmful pollutants contributing to environmental risks is the existence of toxic organic and inorganic contaminants, including phenolic compounds, dyes, agrochemicals, pharmaceuticals, and heavy metals. These contaminants often enter water systems through industrial discharges and agricultural runoff, making them key contributors to water pollution. Even at very low concentrations, these pollutants can inflict serious harm on living organisms, underscoring the urgent need for effective and sustainable treatment solutions [1, 2]. Additionally, a substantial amount of untreated waste containing pharmaceuticals, dyes, and heavy metals is released into the aquatic environment each year, causing significant ecological degradation. Heavy metals such as lead, mercury, cadmium, arsenic, and chromium are especially dangerous, as they can trigger oxidative stress in biological systems, leading to cancer and other severe health problems [3, 4]. Dye pollution alone is estimated to contribute

70,000–200,000 tons of contaminants annually. Organic dyes, a significant pollutant, are frequently discharged into water bodies from diverse industries. These dyes are often categorized by their chemical structures or surface charge, with over 100,000 types produced annually and released through effluents, leading to substantial contamination of lakes, rivers, and ponds [5, 6]. Similarly, antibiotics which are crucial for treating bacterial infections and enhancing public health, are excreted in large amounts through faeces and urine, resulting in environmental contamination. This pollution contributes to the growing issue of antibiotic resistance [7, 8]. Another major pollutant is phenol, a colorless organic pollutant, commonly found in wastewater from industries such as ceramics, leather, petrochemicals, textiles, pharmaceuticals, and others. Its discharge contaminates water, soil, and air, and can cause significant ecological damage in aquatic environments due to its high solubility and toxicity [9]. Many of these pollutants are non-biodegradable and interfere with cellular functions even at low concentrations by penetrating biological membranes [10–12]. According to the United Nations Global Water Development Report, nearly two million tons of industrial, agricultural, and domestic waste, including mining residues, pesticides, fertilizers, and chemical byproducts, are released daily into aquatic systems [13, 14]. This widespread contamination is linked to serious health disorders such as cancer, kidney and liver damage, and skin diseases [15]. Therefore, developing sustainable and efficient wastewater treatment technologies is imperative. A wide range of techniques, including ion exchange [16], precipitation [17], coagulation-flocculation [18], membrane separation [19], adsorption [20], and photocatalytic degradation [21] have been employed for removing pollutants from water. While each offers certain advantages, every method frequently has particular limitations, including high operational expenses and reduced efficiency for particular pollutants, or the risk of generating secondary waste or byproducts that require further treatment. Compared to these methods, adsorption stands out as cost-effective and highly efficient approach. It operates under relatively simple conditions and carries a low risk of producing harmful byproducts or secondary pollutants during the treatment process. The process relies on solid materials with large surface area, abundant active functional groups, and strong chemical and physical stability to capture contaminants from an aqueous system [11].

In recent years, there has been growing interest in carbon-based materials for treating wastewater due to their effectiveness, low cost, and eco-friendly nature. A variety of adsorbents, including activated carbon [22], zeolite [23], clay minerals [24], and biosorbents [25] have been successfully utilized, offering significant adsorption capacity, high selectivity, and the advantage of being regenerated and reused [26, 27]. Activated carbon (AC) is a solid material made of carbon with an amorphous structure, an intricate pore system, and large surface area. It contains various oxygenated functional groups, including carboxylic acids, phenols, carbonyls, and lactones [15, 28]. Activated carbon has gained considerable attention in environmental applications, such as sewage treatment, water purification, air filtration, and greenhouse gas capture [29]. It is widely acknowledged to be efficient at absorbing contaminants, both inorganic and organic pollutants [15, 30, 31]. AC can be distinguished from elemental carbon by various physical and chemical properties [32–34]. According to reports, despite the worldwide manufacturing of activated carbon expanding by approximately 5.5% annually over the past decade, its pace cannot keep up with the rising demand for activated carbon due to population growth. The demand for activated carbon has grown significantly due to the swift social and demographic progress in Asian nations, especially in India and China [35]. It is because a rising population increases the pressure on natural resources and leads to higher levels of pollution. At the same time, rapid industrialization and urbanization in developing countries, such as India and China, have led to increased use of AC in water purification, air treatment, and industrial processes. These combined factors significantly contribute to the growing global demand for AC. However, the substantial expense associated with activated carbon production and regeneration demands affordable materials for AC manufacturing, such as renewable agricultural residues [36].

One of the core principles of green chemistry involves utilizing renewable resources for material synthesis. Biomass, a readily available resource, constitutes the primary renewable source of energy and materials on Earth. This category includes wood, crops, agricultural by-products, food, and more [15, 37]. The Environmental and Energy Study Institute has also listed commonly utilized biomass materials, which are: (1) grain crops and starchy plants, (2) agricultural by-products, (3) food scraps, (4) forestry resources, (5) animal by-products, (6) dedicated energy crops, (7) urban and suburban waste [15]. Therefore, producing activated carbon (AC) from biomass resources has two main benefits such as utilizing renewable resources and reducing reliance on fossil-based feed stocks. While it may contribute to carbon sequestration under certain controlled conditions, the overall carbon footprint depends on the production method. Additionally, unlike biochar, AC is not typically applied to soil without further treatment or purpose-specific modification [38]. Biomass materials exhibit a variety of structures, ranging from periodic pattern to complex hierarchical nanostructures, which make it an excellent precursor for producing activated carbon (AC) with large surface area and well-developed porosity [39–41]. Activated carbon derived

from biomass, often regarded green carbon, is widely utilized in water treatment, environmental remediation, and energy related applications [42–44]. Transforming biomass into high value materials is an efficient waste management approach that contributes to environmental sustainability.

Hydrogels are three-dimensional, water loving polymer networks known for their exceptional water-holding capacity. They have shown promise in water treatment by capturing specific contaminats, depending on their chemical makeup and internal network architecture [45]. These materials are highly adaptable for water purification due to their excellent water adsorption capacity and ability to work in synergy with other treatment methods [46–48]. Chitosan (CS), commonly produced through deacetylation of chitin [49]. It is a naturally occurring alkaline polysaccharide made up of linear chains of β-(1 → 4)-2-acetamido-2-deoxy-D-glucose units [50–52]. It has gained popularity owing to its distinct properties, including affordability, biodegradability, water resistance, non-toxicity, and excellent biocompatibility [53, 54]. Moreover, the presence of amino and hydroxyl groups along the chitosan backbone enables it to bind effectively with colorants and toxic metal ions through electrostatic interactions and hydrogen bonding, making it widely utilized in various wastewater treatment processes [55–57]. Advancements in chitosan-based hydrogels have gained significant attention in biomedical and industrial sectors, encompassing areas like drug delivery [58], tissue engineering [59], and adsorption [60]. Earlier studies also explored chitosan-based films and fibers in biomedical contexts, such as drug delivery and cell adhesion, demonstrating chitosan's biocompatibility and potential for tunable structure and properties. For example, chitosan/polyethylene glycol blend fibers showed promising properties for controlled drug release [61]. In another study, biopolymer films supported pancreatic beta cell adhesion, highlighting the bioactivity of chitosan surfaces [62]. Furthermore, blended films of chitosan with starch and sodium benzoate were evaluated for their structural and antimicrobial features [63]. Thus, the above studies reflect the potential biocompatibility and non-toxicity features of the blends, making them desired candidates for wastewater treatments. The properties of chitosan hydrogels, whether formed through physical or chemical crosslinking, are largely determined by factors like chitosan molecular weight, the nature of the incorporated materials, and the crosslinking method applied [64, 65]. When integrated with biomass-derived activated carbon, these hydrogels benefit from the high adsorption capacity of activated carbon while leveraging chitosan's biodegradability and biocompatibility. This combination results in a composite material with improved efficiency for removing pollutants from water, offering a viable and eco-friendly approach for sustainable water purification.

Moreover, in recent years, researchers have long focused on activated carbon and its extensive applications. Many comprehensive reviews have been published, addressing various aspects of activated carbon, from synthesis methods to its diverse applications [22, 66]. To the best of our knowledge, no comprehensive review has yet focused specifically on the combined use of biomass-derived activated carbon (AC) and chitosan-based hydrogels for water purification. Although each of these materials has been individually studied for contaminant removal, their hybrid integration remains underexplored. Biomass-derived AC offers high surface area and strong adsorption affinity, particularly for organic compounds, while chitosan hydrogels provide excellent biocompatibility, tunable functionality, and high affinity toward metal ions. The synergistic combination of these two materials can result in a multifunctional adsorbent capable of targeting a wide range of pollutants. Therefore, this review is timely and necessary to evaluate and summarize current advancements in this integrated system, offering insights into their fabrication, adsorption mechanisms, performance, and practical applicability in treating complex wastewater matrices containing both organic and inorganic contaminants. Unlike previous reviews, this work provides a unique perspective by emphasizing synergistic and competitive adsorption in multi-pollutant systems, incorporating recent data and highlighting comparative performance metrics of hybrid composites. This review begins with a comprehensive overview of activated carbon and chitosan hydrogel, outlining their key properties, advantages, and formation. The discussion then moves to fabrication techniques used to combine these materials into high performance composites. It examines their applications in remediating pollutants and various factors that influence adsorption efficiency. To provide a clear comparative perspective, we compile a table summarizing recent studies on biomass derived activated carbon combined with chitosan hydrogels, detailing the absorbent materials, synthesis methods, target pollutants, adsorption mechanisms, capacities, and associated isotherms and kinetic models. The review concludes by addressing current challenges and suggesting future perspectives based on the current knowledge, paving the way for new research opportunities. Ultimately, the review provides researchers in gaining a clearer understanding of the current state of biomass-derived activated carbon and chitosan hydrogel composites for the capture of organic and inorganic contaminants. This review also highlights critical knowledge gaps that must be addressed promptly to ensure the effective and sustainable management of these materials for environmental remediation.

## Biomass derived activated carbon/chitosan hydrogels

### Introduction to activated carbon

It is a porous material with an exceptionally large surface area and high reactivity, making it particularly effective for adsorption in water purification applications [67–69]. It is generally produced through the thermochemical treatment of organic precursors at high temperatures in the absence of oxygen or under inert atmospheric conditions. The selection of precursor material for activated carbon production depends on several considerations, including availability, safety, and overall production cost [66]. Its hierarchical pores and surface functional groups enable strong adsorption potential, especially when incorporated into composites for removal of organic pollutants [70, 71]. Its surface area typically varies between 500 and 3000 $m^2/g$. To better understand how synthesis methods influence material characteristics and performance, Table 1 provides a comparative overview of different preparation techniques for activated carbon, detailing their processing parameters, and corresponding adsorption characteristics.

### Biomass-derived activated carbon

Biomass, particularly agricultural and plant-based residues, serves as a valuable precursor for the production of activated carbon due to its abundance, renewability, and high carbon content. Common biomass sources include nutshells, wood waste, coconut shells, and agricultural by-products, which offer a cost-effective and sustainable route for preparing activated carbon used in pollutant adsorption [73, 74]. The composition of biomass varies significantly depending on its origin and source, but it commonly contains carbon-rich components such as cellulose, hemi-cellulose, lignin, and small amounts of proteins and lipids, which make it suitable for activated carbon production. Historically, activated carbon has been produced from coal or charcoal. However, using renewable resources to produce activated carbon is more attractive because of its sustainability. In recent years, the carbonization of naturally occurring grasses and tree leaves has been explored for multiple potential industrial uses, offering a renewable alternative [75, 76]. At the same time, significant progress has been made in tailoring the structure, composition, pore architecture, and morphology of porous carbon-based substances to improve their electrochemical properties. Biomass-derived carbon materials exhibit several advantageous characteristics, including a large surface area, superior electrical conductivity, low density, and strong chemical and thermal stability, as well as cost-effectiveness, indicating their suitability as ideal electrode materials for super capacitors, showcasing their potential in energy storage applications [73]. Natural biomass-derived carbons have gained attention as excellent alternatives to traditional carbon sources for various uses. Agricultural and forestry residues are frequently utilized for producing biomass carbons owing to their economic feasibility and widespread availability. The cost-effectiveness of biomass-derived AC is noteworthy. For instance, the production cost of peanut husk wood apple shell activated carbon (ACNB) is estimated at ₹239.22/kg, significantly lower than commercial activated charcoal (~₹1000/kg), with a 16.7-fold cost advantage over multiple usage cycles, reinforcing its economic feasibility for industrial-scale applications [77, 78]. Every year, activities such as agriculture, logging, and other human operations generate large amounts of residues like rice husks, straws, and wood chips. The high content of lignin and cellulose in these biomass resources makes them a preferred choice for creating cost-effective biomass carbons [79–81]. In terms of the adsorption process, adsorbents made from biomass waste have attracted significant interest due to their affordability, abundant renewable raw materials, and eco-friendliness [79, 82, 83]. However, previous literature research on adsorption frequently assesses various biomass waste-derived activated carbon adsorbents for their ability to capture pollutants [32]. For instance, Das et al. reported the removal of methylene blue utilizing waste rice straw as a biomass-derived activated carbon. The findings revealed an improved uptake ability of 95.24 mg/g for

**Table 1** Comparison of different preparation methods for activated carbon and their influence on physical and chemical properties

| Preparation methods | Activating agents | Temperature range (°C) | Surface area ($m^2/g$) | Pore structure | Functional groups | Advantages | Limitations | References |
|---|---|---|---|---|---|---|---|---|
| Physical activation (2-step) | Steam or $CO_2$ (after carbonization) | 800–1100 | Moderate (400–800) | Mainly micropores | Fewer oxygen groups | Environmentally friendly, good physical power, inexpensive method | Long activation time and low adsorption capacity, and s high energy consumption | [72] |
| Chemical activation | $H_3PO_4$, KOH, $ZnCl_2$ (impregnated before carbonization) | 400- 600 | High (3000) | Micro- & mesopores | Rich oxygen-containing groups | Low energy requirements, low activation time, high microporosity surface | Residual chemicals, corrosion risk | [36] |

MB. The adsorption kinetics followed the pseudo-second-order model, while the equilibrium data closely matched the Langmuir isotherm, suggesting that adsorption occurred as a monolayer on a uniform surface [32]. In another study, Wang et al. investigated the removal of fluoroquinolone antibiotics utilizing hierarchically porous carbon (HPC) derived from abundant straw waste. The HPC exhibited excellent adsorption performance, achieving 99.53% efficiency and exceptional capacity of 199.07 mg/g within 30 min. The adsorption process was best described by the pseudo-second-order kinetic model, and the equilibrium results were consistent with the Freundlich isotherm. The suggested mechanisms behind the adsorption involved nanoscale active sites, hydrogen bonding, and π–π interactions [84]. In summary, biomass-derived activated carbon, sourced from abundant agricultural residues, offers a cost-effective, renewable, and environmentally friendly alternative to traditional carbon materials. Recent advancements have enhanced its structural and adsorption properties, making it highly efficient for pollutant removal; however, adsorption performance varies with biomass type and preparation methods, indicating the need for standardized approaches to optimize its large-scale application.

## Chitosan hydrogels

Chitosan (CS) is a biopolymer of natural origin, produced by partially removing acetyl groups from chitin, which ranks as the second most prevalent natural material. It consists of repeating units of *D*-glucosamine and *N*-acetyl-*D*-glucosamine, which are interconnected by β-1,4 glycosidic bonds [46]. However, hydrogels form 3D-crosslinked structures capable of soaking up and holding substantial quantities of water or other solvents without dissolving, owing to their ability to retain moisture, flexibility, and viscoelastic nature. The polymer chosen for hydrogel formation must possess a specific network structure, including its distinct chemical and aggregate properties, which affect water absorption through the swelling rate [85]. Hydrogels can be synthesized using several techniques, such as grafting, crosslinking, impregnation, chelation, and in situ polymerization [86, 87]. Recently, chitosan-based hydrogels have been effective materials for treating contaminated water because of their excellent mechanical and thermal stability [88], strong chemical resistance, cost-effectiveness, and ease of recovering and reusing both the hydrogel and the pollutants [89–91]. Modified chitosan materials, especially those oxidized, show improved adsorption properties due to the incorporation of functional groups like amino and hydroxyl, which promote stronger binding with organic pollutants. These groups enhance pollutant interaction through mechanisms such as electrostatic attraction and complex formation, offering potential biological relevance in degrading or capturing contaminants. The existence of hydroxyl and amino functional groups in chitosan allows it to effectively adsorb multiple pollutants, including heavy metal ions and dyes primarily through electrostatic interactions and hydrogen bonds. Consequently, integrating the advantageous properties of both hydrogels and chitosan enhances the overall adsorption efficiency and capacity of chitosan-based hydrogel systems [92]. Chitosan-based hydrogels remove pollutants from water through a variety of adsorption mechanisms, like electrostatic attraction, ion exchange, hydrogen bonding, π–π stacking, coordination or chelation, and surface complexation [46, 93]. The development of these hydrogels is closely associated with forming a crosslinked network structure [94, 95]. Hydrogel synthesis methods are classified into two categories based on the crosslinking approach: physical and chemical crosslinking. Physical crosslinking relies on non-covalent forces such as hydrogen bonding, van der Waals interactions, ionic linkages, and hydrophobic effects. In contrast, chemical crosslinking entails the creation of covalent bonds that permanently connect polymer chains [94–96]. These crosslinking methods are crucial for tailoring the hydrogel's properties to enhance its performance in pollutant removal applications. In summary, chitosan-based hydrogels offer a promising platform for pollutant remediation due to their multifunctional adsorption mechanisms and modifiable structure. However, their effectiveness is strongly influenced by the choice of crosslinking methods and functionalization techniques, underscoring the importance of optimizing to achieve consistent and efficient performance in real-world applications.

## Advantages of biomass derived activated carbon and chitosan hydrogels

The biomass-derived activated carbon (AC) and chitosan hydrogels have emerged as sustainable and economical materials with significant potential in environmental cleanup. Activated carbon obtained from biomass utilizes low-cost, renewable sources such as agricultural and forestry waste. This offers high surface area, well developed porosity, and excellent adsorption performance for wide range of pollutants [15, 97]. Moreover, its production aligns with principles of green chemistry, as it reduces waste generation and decreases dependence on non-renewable materials. Figure 1 highlights key features of chitosan hydrogel and biomass-derived activated carbon. On the other hand, chitosan hydrogels, derived from natural chitin, are notable for their biocompatibility, biodegradability, and versatile adsorption mechanisms owing to its rich composition of hydroxyl and amino groups [55, 98]. These hydrogels exhibit good mechanical properties, flexibility, and swelling

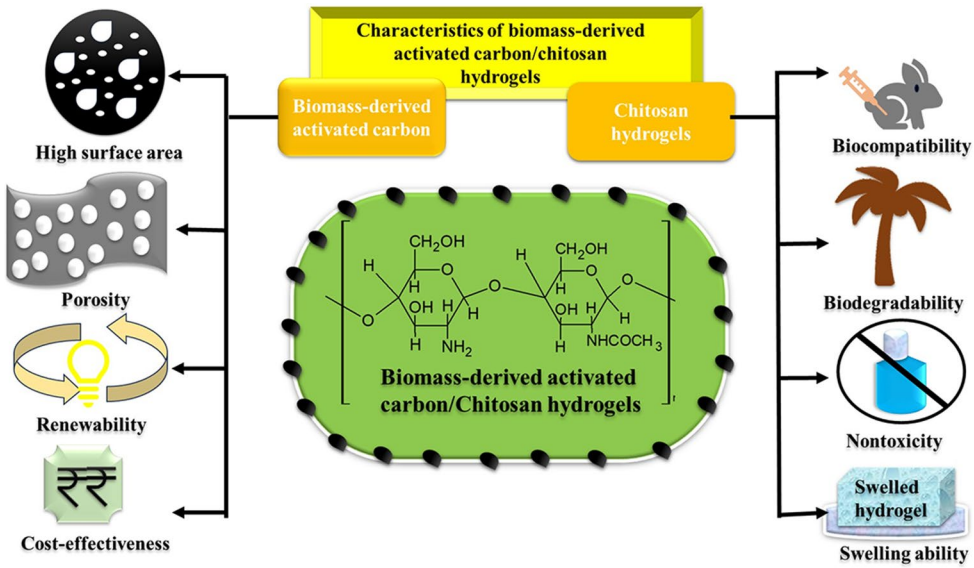



**Fig. 1** Characteristics of biomass-derived activated carbon and chitosan hydrogel

behavior, enhancing their performance in water treatment and controlled-release applications. Their antibacterial properties, environmental safety, and ability to be regenerated and reused further contribute to their practical appeal [95, 99]. Although both chitosan hydrogels and biomass-derived activated carbon offer significant individual benefits, recent studies have focused on combining them into composite systems to enhance pollutant removal. The next section highlights the design strategies, synergistic effects, and environmental applications of activated carbon–chitosan hydrogel composites.

In summary, both biomass-derived activated carbon and chitosan hydrogels demonstrate excellent potential for sustainable pollutant removal due to their eco-friendly natures, high adsorption capacities, and reusability. However, each material has limitations when used independently, which has driven current research toward developing synergistic composite systems for improved efficiency and multifunctionality.

## Activated carbon–chitosan hydrogel composites

The integration of biomass-derived activated carbon into chitosan hydrogels results in a hybrid material that combines the extensive surface area and adsorption capacity of activated carbon with the flexibility, biocompatibility, and functional groups of chitosan. This composite system exhibits enhanced performance for pollutant removal from water due to improved mechanical stability, synergistic adsorption mechanisms, and better reusability. Recent studies have increasingly focused on these composites, highlighting their potential as efficient and sustainable materials for water purification applications. For instance, Zaoui et al. investigated the use of chitosan–activated carbon/metal oxide hydrogel beads for methylene blue and orange G dyes adsorption. The study showed that the material adsorbs exceptional adsorption capacity up to 134.589 mg/g for methylene blue and 323.625 mg/g in case of orange G. The results fit well with the Langmuir isotherm model ($R^2 = 1$), which means the adsorption happened in a single layer on a uniform surface. The process also followed the pseudo-second-order kinetic model, pointing to a heat-absorbing reaction driven by chemical bonding [100]. Jung and colleagues developed composite beads by combining activated carbon and chitosan for removing acid orange 7 (AO) and methylene blue (MB) dye from water. They showed impressive adsorption abilities, capturing up to 511.38 mg/g of AO and 413.08 mg/g of MB. Even after being reused five times, the beads still removed over 82% of AO and 86% of MB. This study shows that chitosan can serve both as a green source for making carbon and as a strong support material, making the beads effective and environmentally friendly [101]. In summary, activated carbon–chitosan hydrogel composites offer a promising multifunctional platform for efficient and reusable pollutant removal, combining the strengths of both materials. However, challenges such as scalability, mechanical durability under real wastewater conditions, and performance consistency over multiple cycles remain areas for further research.

## Fabrication of biomass-derived activated carbon/chitosan hydrogel composites

Biomass residues from diverse sources, including forestry by-products, agricultural leftovers, and agro-industrial refuse, have garnered substantial interest due to their potential to be transformed into porous carbon substances through pyrolysis, hydrothermal, physical, and chemical activations

[102]. Activated carbon is produced in three main steps: biomass pretreatment, carbonization, and activation. Pretreatment removes nutrients and soluble impurities. Carbonization converts lignin and cellulose into carbon-rich material by eliminating water, oxygen, and other elements, typically above 400 °C. This also generates gases like $H_2O$, CO, and $CO_2$, which aid in the activation stage [75]. The process of activation typically involves either physical or chemical approaches [15, 79]. Physical activation involves heating the biomass at high temperatures in an inert gas to develop porous structures. Conversely, chemical activation uses agents like phosphoric acid or potassium hydroxide to enhance porosity. In the physical activation procedure, the carbon-rich substances are heated between 600 and1200 °C along with an activating substance, such as steam, carbon dioxide, oxygen, or sulfur dioxide [103]. Chemical activation of organic biomass is carried out using dehydrating agents like zinc chloride, potassium hydroxide and phosphoric acid. These activating chemicals are mixed with the biomass, followed by thermal treatment at temperatures typically between 400 and 900 °C to develop a porous carbon structure [104]. For instance, Pan et al. fabricated the activated carbon using golden needle mushroom root. Initially, the roots were first cut into small pieces and freeze-dried. The dried pieces were then soaked in a 35% (by wt) phosphoric acid solution for 12 h, followed by drying in an oven for another 12 h. Subsequently, the samples underwent carbonization under a nitrogen atmosphere at 200 °C in the muffle furnace. The resulting carbonized roots were rinsed with distilled water until neutral pH comes, then drying at 60 °C for 6 h and ground into final powdered form. The prepared biomass derived activated carbon exhibited a porous structures attributed to $H_3PO_4$ activation, which likely transformed some micropores into mesopores and enhanced the availability of active sites for dye adsorption [105]. Another study by Benaddi et al. used *Calotropisprocera* fruit for the fabrication of activated carbon. First, collected the fruit and rinsed with distilled water. After cleaning, it was dried in an oven at 108 °C for 48 h. Once dried, it was ground into fine particles and passed through a sieve to obtain a uniform powder. For activation, the prepared powder was impregnated with $H_3PO_4$ at ratios of 1:1 using 1–2 g of powder with corresponding acid volumes. The mixtures were sealed and left at ambient temperature for 24 h, then carbonized at 600 °C under $CO_2$ for 2 h. The activated carbon produced was rinsed until a neutral pH was reached, then dried at 105 °C for 24 h [106]. Once activated carbon is produced, it can be combined with chitosan using methods like impregnation, in-situ polymerization, freeze-drying, crosslinking, and ultrasonication. These techniques enable the development of composites with enhanced adsorption capacity and structural integrity, making them appropriate for water purification. Notably, the incorporation of biomass-derived activated carbon with chitosan hydrogel has demonstrated superior performance over individual components [107].

## Activated carbon entrapped chitosan composites

This approach entails embedding pre-synthesized activated carbon into chitosan matrices using methods such as bead formation, freeze-drying, or ultrasonication. These fabrication techniques physically trap the activated carbon within the hydrogel network, improving structural stability and adsorption performance without undergoing carbonization. In recent years, growing research efforts have explored different preparation strategies on developing activated carbon chitosan hydrogel composites aiming to enhance their performance in environmental applications.

For instance, Kumar et al. developed activated carbon/chitosan hydrogel beads (ACCB) using freeze drying method. The preparation began by dissolving commercial chitosan into an acetic acid solution. Activated carbon was then added and mixture was stirred thoroughly until it became smooth and even. The mixture was gently dropped into a sodium hydroxide solution using a syringe and left to set overnight. The resulting hydrogel beads were then taken out and rinsed with distilled water until the pH matched that of de-ionized water. Finally, the beads were freeze dried and subsequently utilized for further applications. Figure 2a illustrates the fabrication process of activated carbon/chitosan hydrogel beads (ACCB) [108].

Sait et al. utilized the ultrasonication method to fabricate activated carbon/chitosan microbeads. Initially, a chitosan blend was created by mixing chitosan powder in diluted acetic acid and magnetically stirring the mixture for 3 h to form a hydrogel. Next, activated carbon was introduced to the hydrogel, and blend was consistently agitated for 4 h at 60 °C. The prepared hydrogel was gently dripped into a NaOH solution using a micropipette, leading to the formation of hydrogel beads. These beads were thoroughly washed with purified water and drying at 60 °C in oven [110].

## Carbonized composites derived from chitosan and biomass precursors

This class of composites is produced by subjecting chitosan or biomass-based materials to thermal treatment, leading to the formation of porous carbon structures. The carbonization process enhances surface area, stability, and adsorption efficiency, making these materials suitable for advanced water purification applications. Recently numerous studies have reported carbonization methods to develop efficient

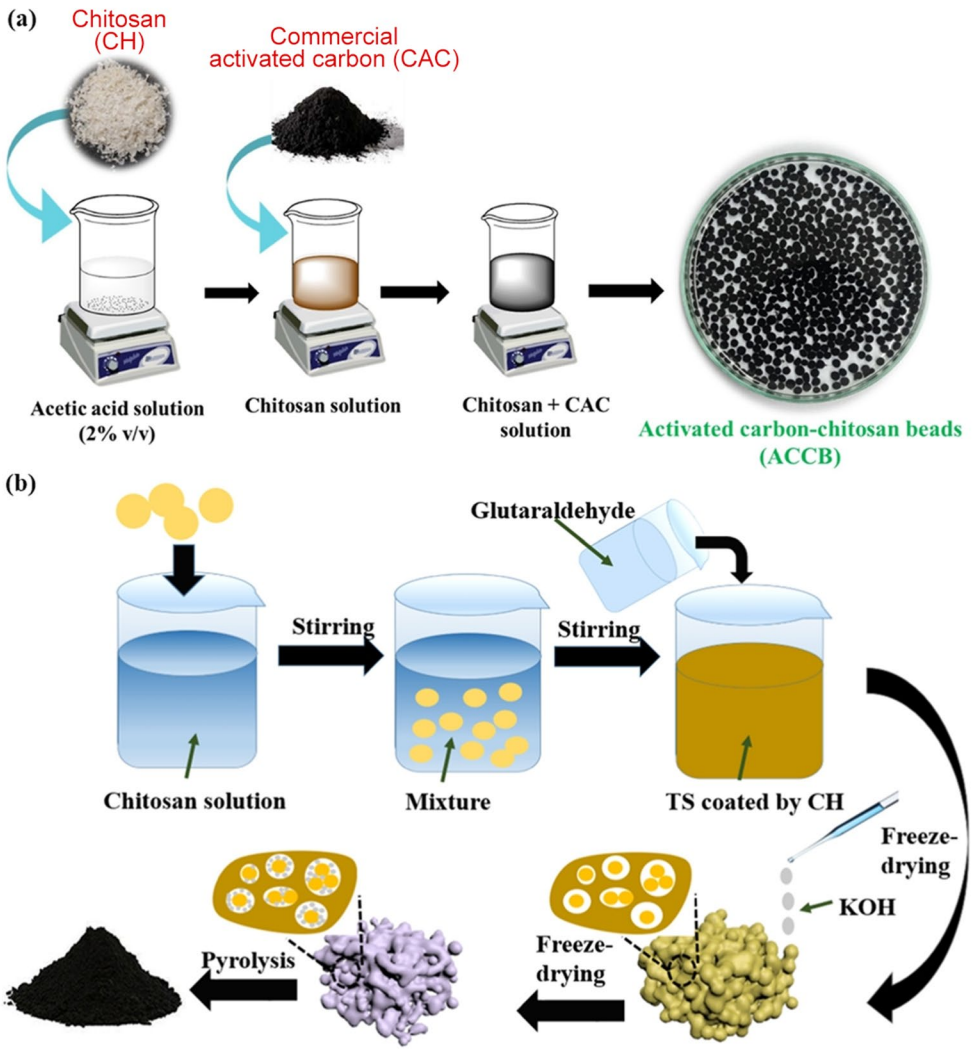



**Fig. 2** **a** Preparation of activated carbon/chitosan hydrogel beads (ACCB) [108], *Reproduced with permission from Elsevier,* and **b** synthesis process of $C_{CH\text{-}TS}$ activated carbon [109], *Reproduced with permission from Elsevier*

chitosan hydrogel activated carbon composites for water treatment applications.

For instance, Ma et al. utilized the freeze-drying method for fabricating novel activated carbon/chitosan hydrogel (CH) coated tea saponin (TS) (illustrated in Fig. 2b). First, a chitosan solution was prepared by dissolving chitosan and acetic acid and stirring for 24 h at 65 °C. Then, TS was introduced into the solution and mixed at 750 r/min. Simultaneously, glutaraldehyde was added, forming a gel that was vacuum freeze-dried for 48 h. Next, the KOH solution in water was introduced into the gel mixture, freeze-dried for about 48 h, and pyrolyzed at 750 °C under a nitrogen atmosphere for 2 h. The resultant material was washed with HCl, followed by sonication and thorough washed with distilled water until pH 7 was achieved. It was then dried in oven for 24 h to obtain $C_{CH\text{-}TS}$ (activated carbon derived from chitosan hydrogel and tea saponin) [109].

A study by Fan et al. developed CS/EDTA/CBC hydrogel using free radical polymerization. Cotton waste was first treated with KOH, sonicated, vacuum-soaked, dried, and carbonized at 400 and 800 °C under nitrogen. The resulting material was thoroughly washed with purified water and nitric acid, and then dried to form cotton biochar carbon (CBC). Subsequently, KOH was neutralized using acrylic acid in an ice-water bath, into which the cotton-based carbon was sonicated. Chitosan and disodium ethylene diaminetetraacetic acid (EDTA-2Na) were then added, followed by ultrasonic treatment with KPS (initiator) and NMBA (crosslinking agent) for 1 h. The mixture was heated for 2 h to form CS/EDTA/CBC hydrogels, which were then washed with ethanol, dried in the oven at 70 °C, and ground for use. Figure 3a depicts the synthesis route for CBC and CS/EDTA/CBC [111].

In another study, Shao et al. synthesized *N*-doped carbon beads derived from lignin and chitosan utilizing an in situ self-assembly approach. First, SL (sodium lignosulfonate) and chitosan mixtures with different ratios were dissolved in de-ionized water, and HCl was added to mix chitosan

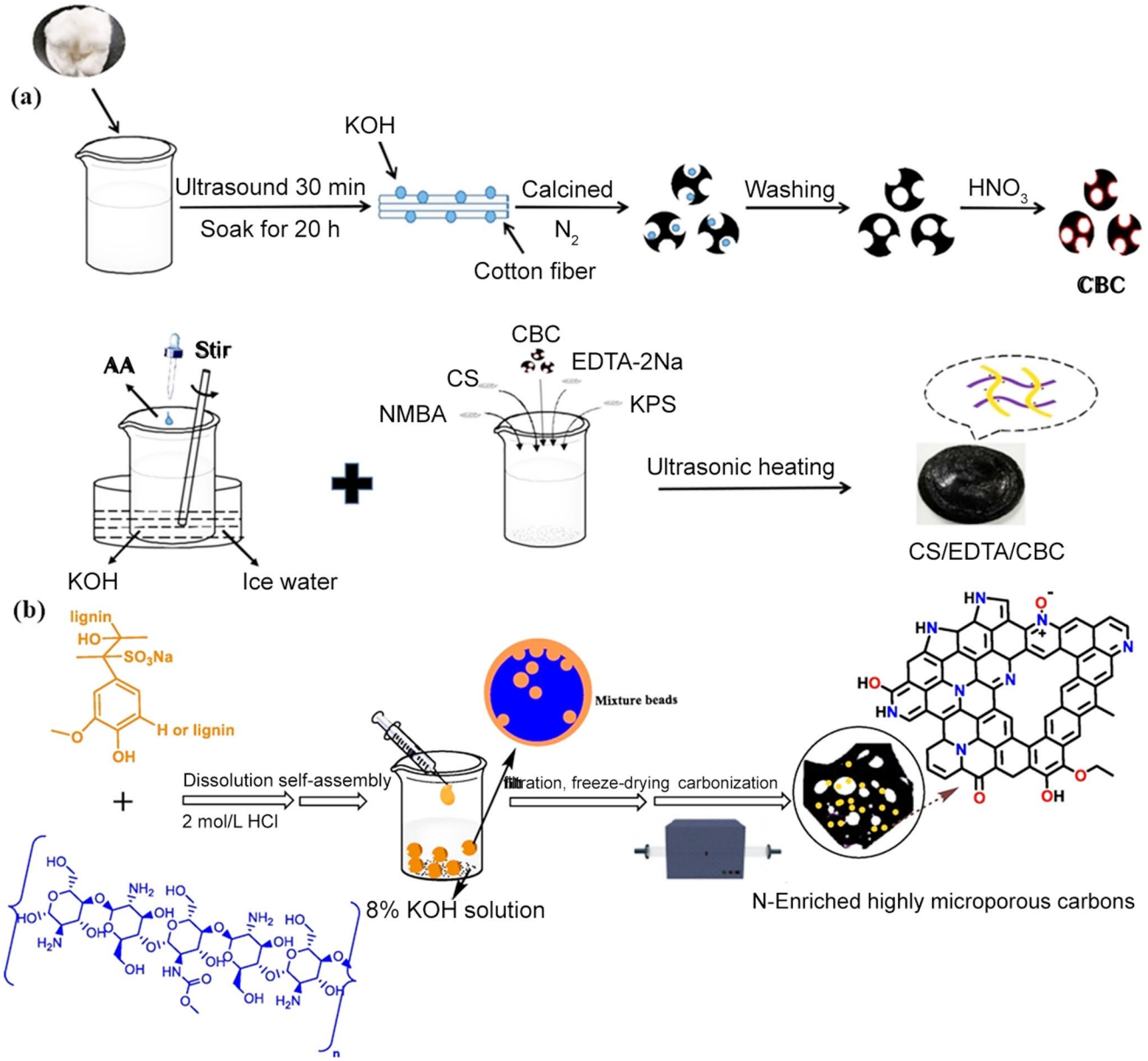



**Fig. 3** **a** Synthesis method for CBC and CS/EDTA/CBC [111]. *Reproduced with permission from Elsevier,* and **b** synthesis process of N-doped highly microporous carbon from lignin/chitosan composite beads [2], *Reproduced with permission from Elsevier*

mixture, forming a solution with SL. This mixture was dipped into KOH solution to form gel beads, which were stabilized for 20 h. The beads were subsequently filtered, freeze-dried, and subjected to carbonization at 600 °C under a nitrogen atmosphere. Following carbonization, the resulting solids were rinsed with hydrochloric acid and deionized water. They were then filtered and dried in an oven at 80 °C for 12 h to produce nitrogen-doped carbon with a highly microporous structure. Figure 3b illustrates the preparation method for this porous carbon, which was synthesized from lignin–chitosan beads [2].

While reviewed studies demonstrate a wide range of fabrication techniques, such as freeze-drying, ultrasonication, in situ polymerization, and chemical activation, their scalability, environmental impact, and industrial feasibility differ significantly. For instance, freeze-drying, though effective in preserving porosity and structural integrity, is energy-intensive and time-consuming, limiting its scalability for industrial use. In contrast, ultrasonication offers a faster and less energy-intensive approach but may compromise uniformity in large-scale batches. Chemical activation using KOH or $H_3PO_4$ is known for producing highly porous carbon, yet the use of corrosive chemicals raises environmental and

disposal concerns, particularly in large-scale production. Similarly, in-situ polymerization offers good dispersion of carbon in the chitosan matrix but often involves complex steps and multiple reagents, reducing process simplicity. Therefore, future studies should not only report synthesis methods but also evaluate cost-effectiveness, regeneration feasibility, and real-wastewater performance, which are critical for industrial translation of these hybrid adsorbents.

## Removal of organic pollutants

### Organic pollutants

In recent years, swift industrial growth and a substantial increase in population have led to severe pollution of water resources. Water contamination has emerged as a pervasive environmental issue, with industrial effluents playing a major role in polluting water bodies [112–114]. The rise in wastewater contamination, combined with the escalating demand for clean water, has significantly burdened economies worldwide [115, 116]. Particularly, the contamination of rivers and surface or groundwater with harmful organic substances like colorants, agrochemicals, synthetic industrial agents, and pharmaceutical and cosmetic products is a serious concern [20, 117–119]. These contaminants are released into the environment in substantial amounts via numerous industrial activities, presenting significant threats to public health [5, 120, 121]. To address these challenges, different techniques like adsorption [122], photocatalytic degradation [123], and membrane filtration [124] have been employed to eliminate these pollutants. Among them, the adsorption method is commonly utilized to eliminate various pollutants due to its simplicity, effectiveness, affordability, and reusability [125, 126]. Figure 4a presents a schematic representation of the adsorption process employed for eliminating organic contaminants, which illustrates the adsorption mechanism of organic pollutants onto biomass-derived activated carbon chitosan hydrogel, highlighting the roles of π–π stacking, hydrogen bonding, and electrostatic attractions facilitated by its porous structure and functional groups. Recently, there have been numerous studies on organic pollutant adsorption using these composites. For instance, Jeyaseelan et al. investigated the removal of CR dye utilizing activated carbon modified chitosan beads. The findings revealed the maximum adsorption of 75%-80% using the prepared beads. The adsorption behavior was consistent with pseudo-first-order kinetics, reflecting a process influenced by the rate of interaction between adsorbent and adsorbate. Additionally, the equilibrium results aligned with the Freundlich isotherm, indicating that the adsorption took place on a non-uniform surface and involved the formation of multiple layers [127]. In a separate investigation, Jung and colleagues explored the removal of methylene blue (MB) and acid orange 7 (AO) dyes using activated carbon embedded within chitosan hydrogel beads. Their findings revealed a significantly improved adsorption performance, with maximum capacities reaching 511.38 mg/g for AO and 413.08 mg/g for MB, highlighting the effectiveness of the composite material in dye removal applications [101]. In summary, while these results demonstrate the efficiency of carbon–chitosan hydrogel composites for dye removal, further investigations are needed to assess their performance in real wastewater containing complex organic mixtures.

### Adsorption method for the elimination of organic contaminants

It is the process in which molecules or ions from a gas or liquid attach themselves to the surface of a solid. In this interaction, the substance being captured is called the adsorbate, while the solid material providing the surface is known as the adsorbent. This surface phenomenon is confined to the outer surface of the adsorbent, with no penetration of the adsorbate into its internal pores or structure [128, 129]. The adsorption process includes two primary types: physical and chemical. Physical adsorption, commonly referred to as physisorption, is governed by relatively weak intermolecular forces such as hydrogen bonding, van der Waals interactions, hydrophobic effects, polarity-induced attractions, dipole–dipole forces, electrostatic interactions, and π–π stacking. These forces allow adsorbate molecules to attach surface without altering their chemical structure. In comparison, chemical adsorption involves the creation of strong chemical bonds between the adsorbate and the adsorbent surface. This typically occurs through electron sharing or transfer, leading to a more stable and energetically favorable attachment [125, 130, 131]. Adsorption is widely regarded as one of the most effective methods for eliminating contaminants from wastewater, and has gained increasing attention due to its numerous advantages [132, 133]. This technique is cost-effective and can be implemented with minimal expense. It is an environmentally friendly approach, helping to minimize harmful impacts on ecosystem. Furthermore, it poses no health risks and does not harm equipment or its surroundings. Additionally, adsorption is non-destructive, preserving the composition of contaminants. These advantages make it a key process in combating water pollution. Its simple design, ease of use, and wide availability often make it the preferred choice over other treatment methods [15]. The sorption efficiency of an adsorbent is affected by several key factors. These include features, like pore size, surface functional groups, morphology, structural features, as well as characteristics of

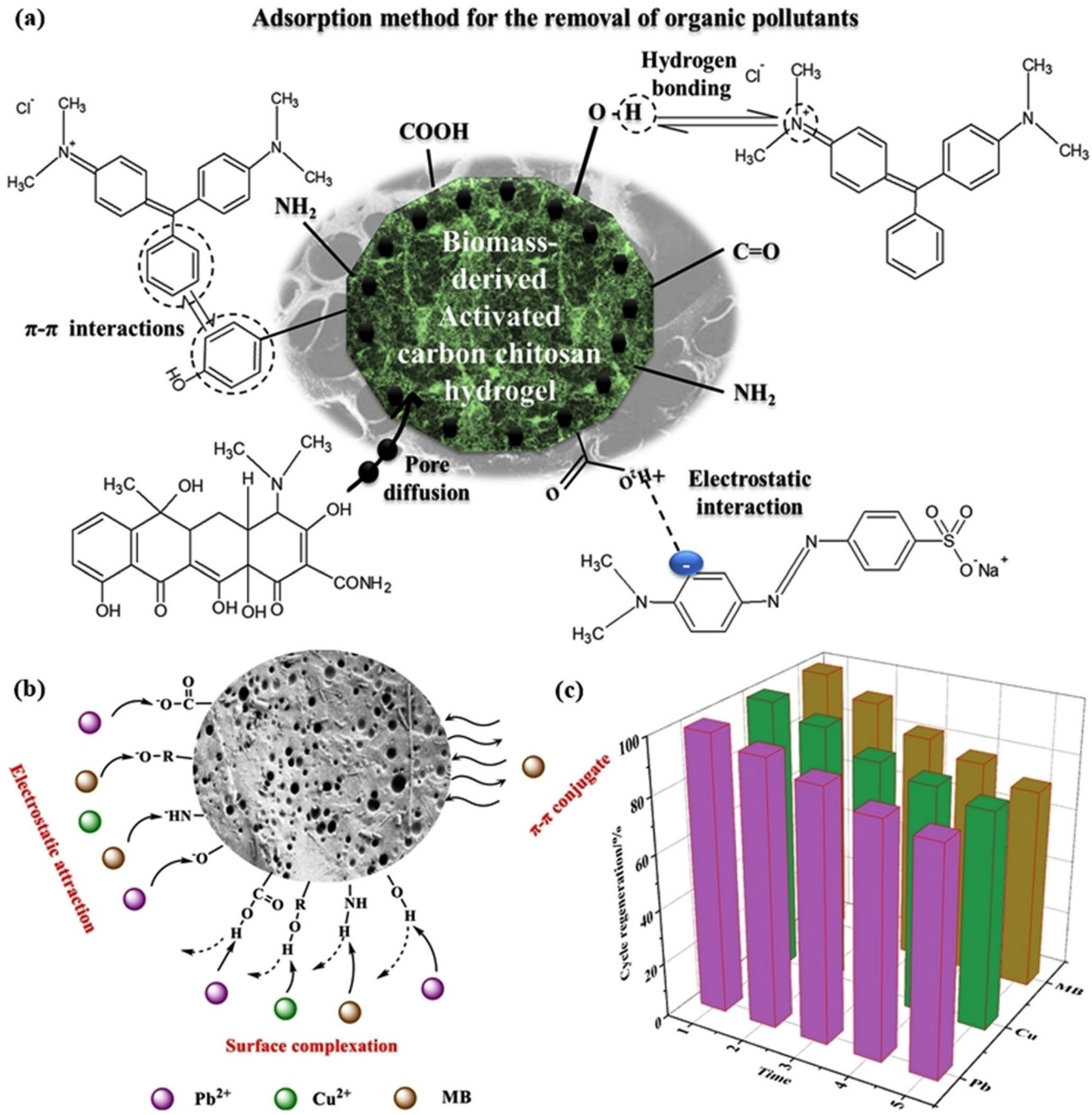



**Fig. 4** **a** Schematic of the adsorption technique used for eliminating organic pollutants, **b** proposed adsorption mechanism for MB using CS/EDTA/CBC hydrogel composite,and **c** cyclic reusability of CS/EDTA/CBC adsorbent [111], *Reproduced with permission from Elsevier*

the adsorbate, including its molecular size, polarity, pKa, molecular weight, and functional groups. In addition, operational parameters like temperature, pH, contact time, and ionic strength significantly impact the adsorption behavior. The combined influence of these elements ultimately governs the efficiency and performance of the adsorption process across different applications. Despite their potential, many adsorbents face limitations due to complex synthesis procedures, high costs, instability, and low hydraulic conductivity. In contrast, activated carbon has demonstrated remarkable efficiency in eliminating contaminants from wastewater, offering a practical solution [134, 135]. Moreover, chitosan-based porous carbon materials derived from renewable sources like agricultural, agro-industrial, and forestry biomass provide versatile and eco-friendly alternatives for water treatment and environmental remediation [136].

Recent research has focused on using biomass-derived activated carbon/chitosan hydrogels for adsorbing organic contaminants. Moreover, comparative analysis has recently used biomass-derived activated carbon/chitosan hydrogels for removing organic pollutants through the adsorption technique, outlined below and concisely presented in Table 2.

For example, Fan et al. investigated the removal of methylene blue (MB) dye utilizing functionalized cotton charcoal/chitosan hydrogel composite (CS/EDTA/CBC). The findings showed an improved uptake capacity of 590.72 mg/g. The adsorption mechanism, as depicted in Fig. 4b, revealed a monolayer adsorption pathway, where the rate-controlling step was identified as a chemical chelation reaction. XPS analysis further demonstrated that surface complexation and electrostatic interactions played dominant roles in facilitating the adsorption process. Additionally, the regeneration study (as depicted in Fig. 4c demonstrated that CS/EDTA/CBC retains 73% adsorption efficiency for methylene blue even after five cycles [111].

Jung et al. conducted a study in which they developed hydrogel beads composed of chitosan and activated carbon (C-AC/ChB). These composite beads were effectively employed to adsorb and remove two synthetic dyes methylene blue and acid orange 7 from contaminated water. The findings demonstrated that the prepared hydrogel exhibited effective adsorption capacities for MB (413.08 mg/g) and AO (511.38 mg/g). Additionally, the adsorption mechanisms for these two organic pollutants were investigated, revealing that the main mechanisms for dye removal in composite involve two key processes: first, the physical trapping of dye molecules within the tiny pores of the chitosan-derived activated carbon; and second, the strong affinities between the dye molecules and the reactive functional groups present on the surface of the C-AC/ChB composite material. These interactions were largely attributed to the presence of surface functional groups such as hydroxyl, carboxyl, and amino groups, which played a key role in capturing dye molecules. Moreover, analysis through XPS revealed changes in surface chemical composition which contribute to improving dye adsorption efficiency [101].

Shao et al. investigated the adsorption of *para*-nitrophenol (PNP) using nitrogen-rich, highly microporous carbon material derived from self-assembled lignin/chitosan composite beads. The material exhibited an impressive uptake capacity of 592.8 mg/g for PNP. Isotherms and kinetics analysis suggested that the adsorption occurred in multilayer pattern on heterogeneous surfaces, with chemisorption being the primary mechanism. The removal process was facilitated by combination of interactions, including hydrogen bonding, π–π stacking, electrostatic forces, acid–base interactions, and micropore filling. Together these mechanisms contributed to the effective removal of PNP using lignin/chitosan-based adsorbent [2].

Zaoui et al. developed chitosan-activated carbon/metal oxide hydrogel beads for the removal of Orange G (OG) and methylene blue (MB) dye from aqueous solution. The results demonstrated significant adsorption capacities, reaching 323.625 and 134.589 mg/g for OG and MB. The adsorption behaviour of the hydrogel beads was best described by the Langmuir isotherm model, with correlation coefficients ($R^2$) close to 1 indicating a uniform and monolayer adsorption process. Additionally kinetic studies that the data closely followed pseudo-second-order model, suggesting that adsorption was primarily driven by chemical interactions between the adsorbent and the dye molecules. The process was also found to be endothermic, suggesting that higher temperatures favored the adsorption efficiency. Additionally, the reusability of the hydrogel beads was evaluated over five consecutive cycles. A slight decline in removal efficiency was observed, with a reduction of 27%–32% for methylene blue and 33%–36% for Orange G, demonstrating the material's potential for repeated use with acceptable performance loss [100].

Another study by Ma et al. investigated the removal of methylene blue dye utilizing activated carbon chitosan hydrogel-coated tea saponin. The findings revealed an enhanced uptake of 604.10 mg/g. The adsorption mechanism of the CCH-TS composite (chitosan hydrogel and tea saponin-derived activated carbon), was found to be governed by multiple interaction forces. These included hydrogen bonding, electrostatic attractions, van der Waals forces, and π–π stacking, all contributing to the effective capture of pollutants. The equilibrium adsorption data closely matched with the Langmuir isotherm model, suggesting monolayer adsorption on a uniform surface. In addition, kinetic analysis aligned well with the pseudo-second-order model, implying that adsorption process was primarily governed by chemical interactions [109].

Sait et al. developed novel activated carbon/chitosan hydrogel microbeads for the adsorption of direct blue 78 dye (DB78) dye from wastewater. The study reported an impressive uptake ability of 32.8 mg/g for eliminating DB78. Adsorption data closely followed Langmuir isotherm model, indicated by an $R^2$ value of 0.99, and kinetics were best described pseudo-first-order kinetic model. These results highlight the high efficiency of activated carbon/chitosan microbeads in removing direct blue 78 dye, presenting a promising strategy for effective dye removal from wastewater [110].

Kumar et al. utilized activated carbon-chitosan beads (ACCB) as an adsorbent to remove antibiotic diclofenac from water. The study showed a significantly high adsorption capacity, with the ACCB achieving a diclofenac (DIF)

**Table 2** Adsorption of various organic and inorganic contaminants using biomass-derived activated carbon/chitosan hydrogels

| Absorbent | Synthesis method | Pollutant | Adsorption mechanism | Adsorption capacity (mg/g) | Isotherms and kinetics | References |
|---|---|---|---|---|---|---|
| Chitosan-derived activated carbon/chitosan composite beads | Simple dripping method | Methylene blue<br>Acid orange 7 | Physical adsorption and electrostatic interaction | 413.08<br>511.38 | Langmuir isotherm and pseudo-second-order kinetics | [101] |
| Carbon derived from the self-assembled lignin/chitosan beads | Self-assembly and one-step activated carbonization method | p-nitrophenol | Hydrogen bond, π-π interaction, electrostatic interaction, and, acid–base interaction | 592.8 | Langmuir model and Pseudo second-order kinetics | [2] |
| Cotton charcoal/chitosan biomass-based hydrogel | Free radical polymerization | Methylene blue | Surface complexation and electrostatic attraction | 590.72 | Langmuir adsorption isotherm and pseudo-second-order kinetics | [111] |
| Chitosan-coated biomass-based magsorbent beads | Free radical polymerization- | Methylene blue (MB) | – | 102.04 | Langmuir isotherm and pseudo-second-order kinetic model | [137] |
| Date palm seed-derived activated carbon/chitosan hydrogel microbeads | Ultrasonication method | Direct blue 78 | Electrostatic interaction | 32. 8 | Pseudo-first-order kinetics and Langmuir isotherm model | [110] |
| CSAC and CHS-CSAC | – | Methylene blue (MB)<br>Methyl orange (MO)<br>Crystal violet (CV) | Intra-particle diffusion | 217.35 (CSAC)<br>105.2 (CHS-CSAC)<br>320.5 (CSAC) | Pseudo-second-order and Elovich model kinetics, Langmuir isotherm model | [138] |
| Chitosan/activated carbon composite beads | Ultrasonication method | Caffeine | Chemical adsorption | 121.90 | Langmuir isotherm and Elovich model | [139] |
| Activated carbon/chitosan hydrogel coated with tea saponin | Freeze-drying method | Methylene blue | Electrostatic attraction, van der Waals forces, H-bonding, π-π interactions, intra-particle diffusion, and pore filling | 604.10 | Pseudo-second-order kinetic model and Langmuir isotherm | [109] |
| Activated carbon/chitosan hydrogel beads | – | Diclofenac | Chemisorption | 99.29 | Freundlich isotherm and pseudo-second-order kinetics | [108] |
| Magnetically retrievable $BaFe_{12}O_{19}$-activated charcoal-chitosan composite hydrogel | Ultra-sonication assisted | Methyl blue | – | 254.6 | Pseudo-second-order kinetic model, Redlich-Peterson as well as Sips adsorption isotherm | [140] |
| Leached carbon black waste and chitosan-carbon hydrogel beads | Gelation method | Tetracycline<br>Amoxicillin | Intraparticle diffusion | 39<br>12 | Pseudo 2nd-order model | [141] |
| Magnetic $Fe_3O_4$@chitosan carbon microbeads | Gelation-thermal cracking process | Doxycycline | – | 4.816 | – | [142] |
| Chitosan/activated carbon composite hydrogels | Wet-casting process | Phenol | Surface complexation and pore diffusion | 41 | – | [143] |
| Chitosan/activated carbon hydrogels | – | Food blue2<br>Foodred17 | – | 1428<br>1370 | Langmuir model | [144] |
| Activated carbon/chitosan hydrogel beads | – | Reactive Black 5<br>Basic Violet 10 | – | 639.8<br>50.7 | Langmuir model | [145] |
| Chitosan/activated carbon/montmorillonite hydrogel beads | Dispersion and coagulation method | Metoprolol | Hydrophobic interactions and Van der Waals interactions | 26 | – | [146] |
| Magnetic-activated carbon/chitosan beads | Cross-linking and freeze-drying method | Ni (II) ions | Ion exchange, complexation and electrostatic attraction | 108.70 | Pseudo-second-order kinetics and Langmuir isotherm model | [147] |

**Table 2** (continued)

| Absorbent | Synthesis method | Pollutant | Adsorption mechanism | Adsorption capacity (mg/g) | Isotherms and kinetics | References |
|---|---|---|---|---|---|---|
| Cotton charcoal/chitosan biomass-based hydrogel | Free radical polymerization | $Pb^{2+}$<br>$Cu^{2+}$ | Surface complexation and electrostatic attraction | 1105.78<br>678.04 | Langmuir isotherm model and pseudo-second-order kinetics | [111] |
| Activated carbon coated with chitosan hydrogel | – | Palladium (Pd)<br>Platinum (Pt) | – | 43.48<br>52.63 | Langmuir isotherm model and pseudo second order kinetics | [148] |
| Chitosan gel-coated activated carbon | – | Fe | – | 2.076412 | Langmuir adsorption isotherm | [149] |
| Chitosan-activated carbon/zinc oxide hydrogel beads | – | $Cd^{2+}$ | Physisorption and chemisorption | 18.63 | Pseudo-second-order kinetics and Freundlich, Redlich-Peterson isotherms | [150] |
| Iron-modified chitosan/coconut shell-activated carbon composite beads | – | Cr(VI) | Electrostatic attraction and complexation | 64.49 | Langmuir model and pseudo-second-order kinetic | [151] |
| *Terminalia catappa* fruit shell-derived activated carbon/chitosan hydrogel- | Chelation method | Cr(VI) | Physical adsorption and hydrogen bonding | 184.45 | Langmuir isotherm | [152] |
| Chitosan-activated carbon@metal oxide hydrogels beads | Cross linking method | Orange G<br>Methylene blue | Hydrogen bonding, Van Der Waals interactions, pore fitting and π–π interactions | 323.625<br>134.589 | Langmuir isotherm and pseudo-second order model | [100] |
| Activated algae/ chitosan/ carboxymethyl cellulose hydrogel beads | Cross linking method | Ni(II) | Hydrogen bonding and electrostatic attractions | 301.05 | Langmuir isotherm and pseudo-second order model | [153] |
| Activated algae/MOFs/chitosan hydrogel beads | Cross-linking method | Co(II) | Electrostatic interactions, complexation, and diffusion | 445.8 | Pseudo-second-order model and the Langmuir isotherm | [154] |

uptake of 99.29 mg/g, reflecting its strong binding energy. In addition, the adsorption mechanism of diclofenac (DIF) onto ACCB was examined, as illustrated in Fig. 5a. The analysis revealed that at pH 6.0, DIF molecules predominantly exist in a positively charged form, enabling robust electrostatic interactions with the negatively charged surface of ACCB. Some DIF molecules form hydrogen bonds through polar groups like –OH and $–NH_2$, while weak π–π interactions take place between the aromatic rings present in the DIF structure and those on the ACCB surface. The reusability of ACCB was evaluated through multiple adsorption–desorption cycles, as shown in Fig. 5b. The results indicated a progressive decline in adsorption efficiency with each successive use, with notable retention of performance (> 50%) observed only during the initial three cycles. The removal efficiency of DIF decreased from 71% in the first cycle to just 36% by the fifth, primarily attributed to surface deterioration and pore obstruction. These structural changes likely reduced the number of accessible active sites, thereby impairing the material's adsorption capacity. Additionally, the influence of different concentrations of NaCl on the adsorption behavior of diclofenac (DIF) onto the ACCB adsorbent was examined, as illustrated in Fig. 5c. As NaCl concentration rises, DIF removal diminishes. Higher ionic strength weakens the electrostatic interaction b/w oppositely charged adsorbate and adsorbent, causing particle aggregation and reducing available functional sites, thereby lowering adsorption capacities [108].

In summary, this section highlights the efficiency of different adsorbent materials in eliminating organic contaminants from wastewater. Adsorption, a process where contaminants adhere to a solid surface, is noted for its simplicity, cost-effectiveness, and minimal environmental impact. Adsorbents like biomass-derived activated carbon/chitosan hydrogel composites demonstrate substantial efficiencies. Notably, activated carbon/chitosan hydrogel has exhibited high adsorption capacities, with methylene blue dye achieving up to 604.10 mg/g using chitosan-coated tea saponin-based activated carbon. In contrast, the lowest adsorption capacity was 32.8 mg/g for Direct Blue 78 dye using activated carbon/chitosan microbeads investigated. When benchmarked against other bio-derived and conventional activated carbons, these composites demonstrate notable advantages. For example, granular activated carbon from coal, as studied by Xie et al., demonstrated exceptional uptake capacity of 169.91 mg/g for phenol due to its well-developed micro/mesoporous structure [155]. Similarly, Maguana et al. reported a capacity of 260 mg/g for methyl

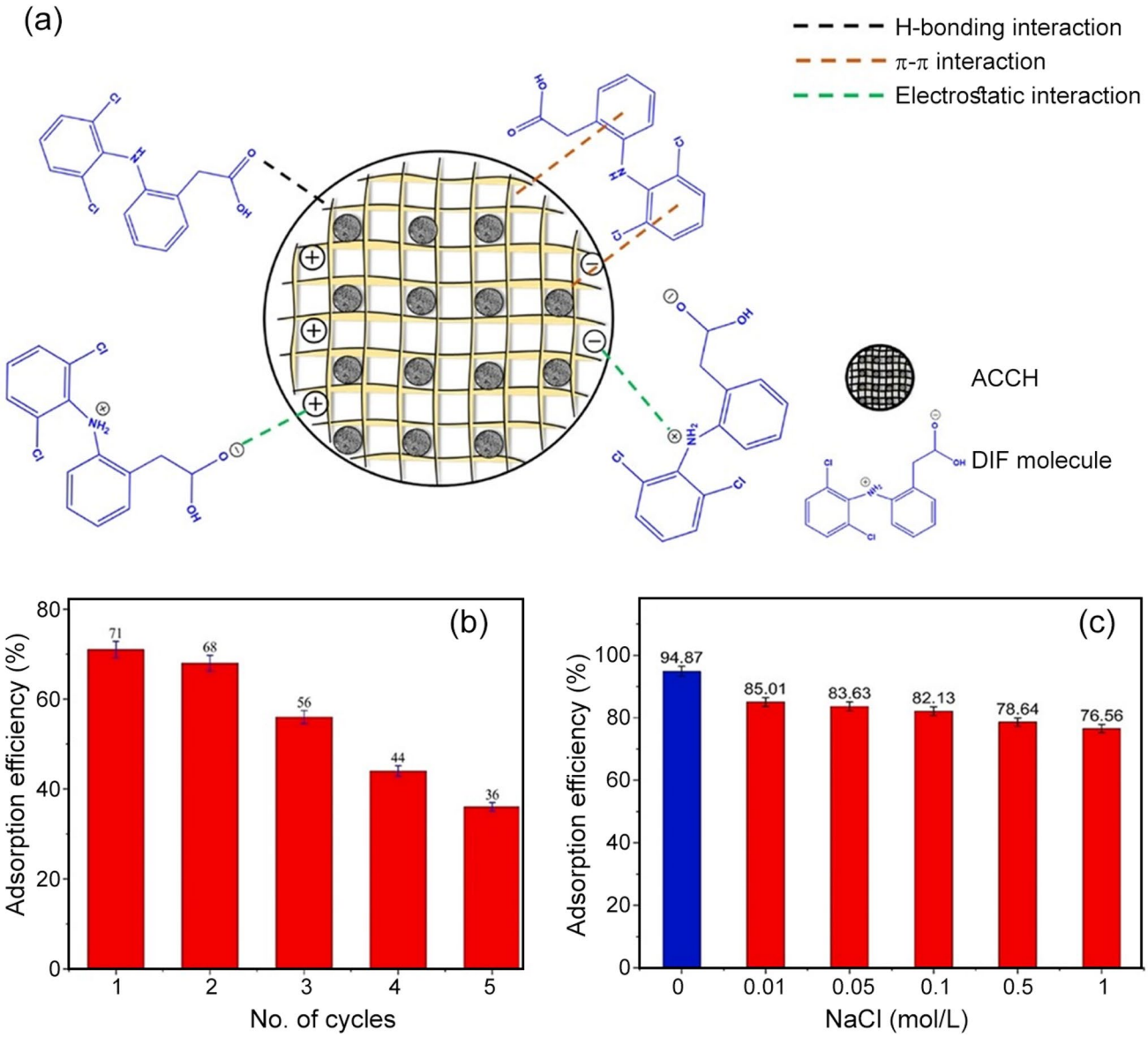



**Fig. 5** **a** Proposed mechanism for DIF adsorption onto ACCB, **b** desorption and reusability analysis of ACCB, and **c**impact of NaCl ionic strength on DIF adsorption [108], *Reproduced with permission from Elsevier*

orange using activated carbon derived from prickly pear seed cake [156]. However, a comparatively lower uptake capacity of 5.6 mg/g was observed for Rhodamine B using corn stalk-derived activated carbon [157]. This variation further supports the superior performance of activated carbon chitosan hydrogel composites over activated carbons. Moreover, the adsorption process is governed by multiple mechanisms, including electrostatic interactions, hydrogen bonding, and π–π stacking. The application of kinetic and isotherm models, specifically, the pseudo-second-order kinetics and the Freundlich isotherm further highlights the system's complex characteristics, indicating a heterogeneous surface and multilayer adsorption behavior.

Table 2 presents the comparative overview of biomass derived activated carbon/chitosan hydrogel-based adsorbents, highlighting their synthesis methods, targeted pollutants, adsorption capacities, mechanisms and kinetic models. Among the listed adsorbents, activated carbon chitosan hydrogel composites exhibited the highest adsorption capacity of 1428 mg/g for food blue -2 and 1370 mg/g for food red 17 dyes, attributed to multiple mechanisms including electrostatic attraction, π–π interactions, and pore filling. Similarly, cotton charcoal/chitosan hydrogels demonstrated high adsorption capacity for heavy metals (e.g., $Pb^{2+}$ 1105.78 mg/g), owing to surface complexation and electrostatic attraction. The ultrasonication and free radical polymerization methods appear to be particularly effective

in producing porous structures with enhanced adsorption performance. In contrast, certain adsorbents like date palm seed-derived activated carbon/chitosan hydrogel microbeads, chitosan-carbon hydrogel beads and chitosan-activated carbon/zinc oxide hydrogel beads showed relatively low capacities, suggesting limitations in surface area or functional group availability. Most of the materials exhibited adsorption behavior consistent with the Langmuir isotherm model, suggesting that adsorption occurred primarily as a monolayer on uniform surface sites. Additionally, the adsorption kinetics were best described by the pseudo-second-order model, indicating that chemisorption played a dominant role in the process. Beyond these primary mechanisms, other interactions—such as hydrogen bonding, acid–base interactions, and intraparticle diffusion also contributed significantly to enhancing the overall adsorption efficiency. Overall, the data suggest that combining activated carbon with chitosan in hydrogel form not only improves structural stability and reusability but also enhances adsorption efficiency through synergistic interactions, especially when functionalized or magnetized.

## Removal of oil pollutants from seawater using chitosan hydrogels

Water resources are significantly polluted by oil-containing wastewater discharged during natural gas extraction and mining activities. Additionally, oil spills that occur throughout the processes of extraction, refining, and transportation of oil further contribute to environmental contamination, posing serious threats to aquatic ecosystems and water quality. This pollution also poses serious threats to both environmental ecosystems and human well-being [158]. Therefore, managing the discharge of industrial oily wastewater has become a critical obstacle in the domain of water purification. In this context, hydrogels have emerged as a promising biomaterial for oil/water separation due to their outstanding water absorption ability, superleophobic nature, high filtration efficiency, and cost-effectiveness [55]. Hydrogels have attracted attention for their potential to overcome the challenges associated with separating oil from water. Among various types, chitosan-based hydrogels have shown great potential. Chitosan, derived through the deacetylation of chitin, ranks as the second most prevalent natural polymer following cellulose. Its molecular structure is characterized by a high density of amino and hydroxyl groups, which contribute to its versatile chemical reactivity and broad applicability. It exhibits excellent properties such as being biodegradable, biocompatible, antibacterial, and easily modified. Additionally, it has a strong ability to bind with metal ions. Due to these properties, chitosan is widely utilized in areas like biomedicine and environmental monitoring. When introduced into hydrogel structures, it not only improves biocompatibility but also imparts effective antibacterial properties and ion exchange capabilities [95]. Chitosan-based hydrogels have emerged as highly effective materials for treating polluted water, owing to their mechanical strength, thermal stability, high chemical resistance, and cost effectiveness. One of their key advantage lies in their ease of recovery and reuse which not only supports the sustainable practices but also allows for potential recovery of adsorbed pollutants [46]. These features make them especially valuable for real world water purification applications. Recent research has highlighted the promising role of chitosan-based hydrogels in addressing oil pollution in marine environments. For example, Kordjazi and colleagues used super-hydrophilic and oleophobic chitosan/acrylamide hydrogel prepared using an in-situ synthesis method. The prepared hydrogel exhibits exceptional performance in separating oil from oil–water mixtures, achieving by 98%. The hydrogel was integrated onto cotton fabric and characterized through FTIR and SEM analyses, which confirmed the successful formation of a porous structure. Swelling studies further revealed excellent water retention and demonstrated strong stability in a seawater-simulated environment [159]. In another study, Saruchi et al. investigated the elimination of crude oil using chitosan-based hydrogel and reported impressive adsorption efficiency of 97%. The hydrogel performed best under acidic conditions, followed by alkaline, with the lowest oil uptake observed at neutral pH. Adsorption studies showed that the data closely followed Langmuir isotherm model suggesting monolayer adsorption on a uniform surface. Furthermore, the kinetic data aligned well with the pseudo-second-order model, indicating that the chemical interactions played a dominant role in the adsorption process. These findings highlight the hydrogel's strong potential for effectively removing oil from contaminated water [160].

## Factors affecting the adsorption performance

Understanding the factors influencing the adsorption efficiency of biomass-derived activated carbon/chitosan hydrogel composites is crucial for optimizing the efficiency of removing organic pollutants. The effectiveness of these materials is highly dependent on various environmental conditions and the type of contaminants. Here, we will delve into the five key factors: pH, temperature, pollutant concentration, dosage, ionic strength, and how they affect the adsorption process. All these factors are summarized in the following subsections.

### pH

The solution's pH significantly influences the adsorption process by directly altering the surface charge of adsorbent, the reactivity of its functional groups, and interaction with co-existing ions. It also impacts the properties of both the adsorbent and the adsorbed molecules, including their dissociation behavior and chemical structure. Variation in pH can influence the adsorption mechanism and ionization of dye molecules, impacting the entire effectiveness of the process [161–163]. In recent years, research has explored how various biomass-derived activated carbon/chitosan hydrogel adsorbents perform against organic pollutants, focusing on the effect of solution pH. For instance, Jung et al. investigated the impact of pH onto C-AC/ChB hydrogel. As depicted in Fig. 6a and b, the adsorption experiments for AO and MB dyes were conducted at multiple pH levels (3, 5, 7, and 10). AO exhibited maximum adsorption at an acidic pH of 3, reaching 290.7 mg/g, whereas MB demonstrated superior uptake at a basic pH of 10, with a removal capacity of 270.9 mg/g. This difference in performance is attributed to large surface area of the activated carbon, which facilitates dye uptake through its intricate pore structure. This process is governed by multiple interaction mechanisms, including Van der Waals forces, π–π stacking, hydrogen bonding, and electrostatic interactions that vary with pH [101].

In another study, Sait et al. examined the effect of pH on removing Direct blue 78 (DB78) dye. The findings revealed that for $AC_{400°C}$ MB-0.3:1, maximum adsorption effectiveness (97%) occurred at pH 5, while under alkaline conditions (pH 9), it was 92%. Similarly, for $AC_{1000°C}$ MB-0.3:1 had its highest performance (96.8%) at pH 4, but only 80% at pH 9. The better performance in acidic conditions is due to the abundance of hydrogen ions ($H^+$), which enhance the positive surface charge of activated carbon, increasing its attraction to the negatively charged dye molecules [110].

### Temperature

Temperature is a crucial parameter influencing removal performance of activated carbon chitosan hydrogel composites. As the temperature increases, the adsorbent's capacity to uptake pollutants improves due to the enhanced kinetic energy, which accelerates the adsorption process [164–166]. The solution temperature can influence the solid–liquid system in two ways: (i) altering the adsorption rate; and (ii) modifying the adsorption capacity. An elevation in temperature can enhance the movement of adsorbate species and accelerate the intraparticle diffusion rate toward the pores on the adsorbent surface. This effect is due to the reduction in the solution's viscosity, which boosts the adsorbate's mobility and lessens the mass transfer resistance to the adsorbent surface [167, 168]. Recent studies have explored the effectiveness of biomass-derived activated carbon/chitosan hydrogel adsorbents for removing organic contaminants, with a focus on how temperature affects their efficiency. For instance, Ma et al. investigated the effect of temperature on MB adsorption using activated carbon/chitosan hydrogel-coated tea saponin. As depicted in Fig. 6c, a minor reduction in adsorption capacity was noted, decreasing from 199.3 to 197.5 mg/g, along with a slight drop in removal efficiency from 99.7 to 98.7%. These results imply that the methylene blue adsorption onto CCH-TS follows an exothermic process [109]. These findings confirm that temperature changes can influence the adsorption mechanism and performance of activated carbon/chitosan composites in pollutant removal.

### Pollutant concentration

The initial concentration of contaminants primarily influences adsorption efficiency of activated carbon chitosan composites by creating a stronger driving force. Higher initial concentrations can accelerate the removal of contaminants by enhancing the interaction between pollutants and active sites available on the surface of composite material [166, 169]. However, increasing the initial adsorbate concentration reduces the removal percentage, although the number of contaminants captured per unit weight of the adsorbent rises. This increase is attributed to stronger driving forces that help overcome the mass transfer resistance of phenolic compounds between the liquid and solid phases. This enhances the frequency of collisions between the adsorbate molecules and the sorbent surface, thereby improving adsorption efficiency [170, 171]. Moreover, as the initial dye concentration increases, the adsorption rate often declines due to the saturation of available binding sites on the adsorbent surface, limiting further uptake of dye molecules [172].

For instance, Fan et al. studied the influence of pollutant concentration using cotton charcoal/chitosan biomass-based hydrogel (CS/EDTA/CBC) adsorbent for adsorbing MB dye (depicted in Fig. 6d). As initial concentration of pollutants rises, the uptake ability of CS/EDTA/CBC initially rises before reaching equilibrium. Furthermore, higher concentrations create a greater conc. gradient across the adsorbent surface, enhancing pollutant binding to adsorption sites. However, with a fixed amount of adsorbent, the adsorption sites are limited, leading to equilibrium once all sites are occupied [111].

### Adsorbent amount

The quantity of adsorbent applied plays a crucial role in determining the adsorption performance of composites

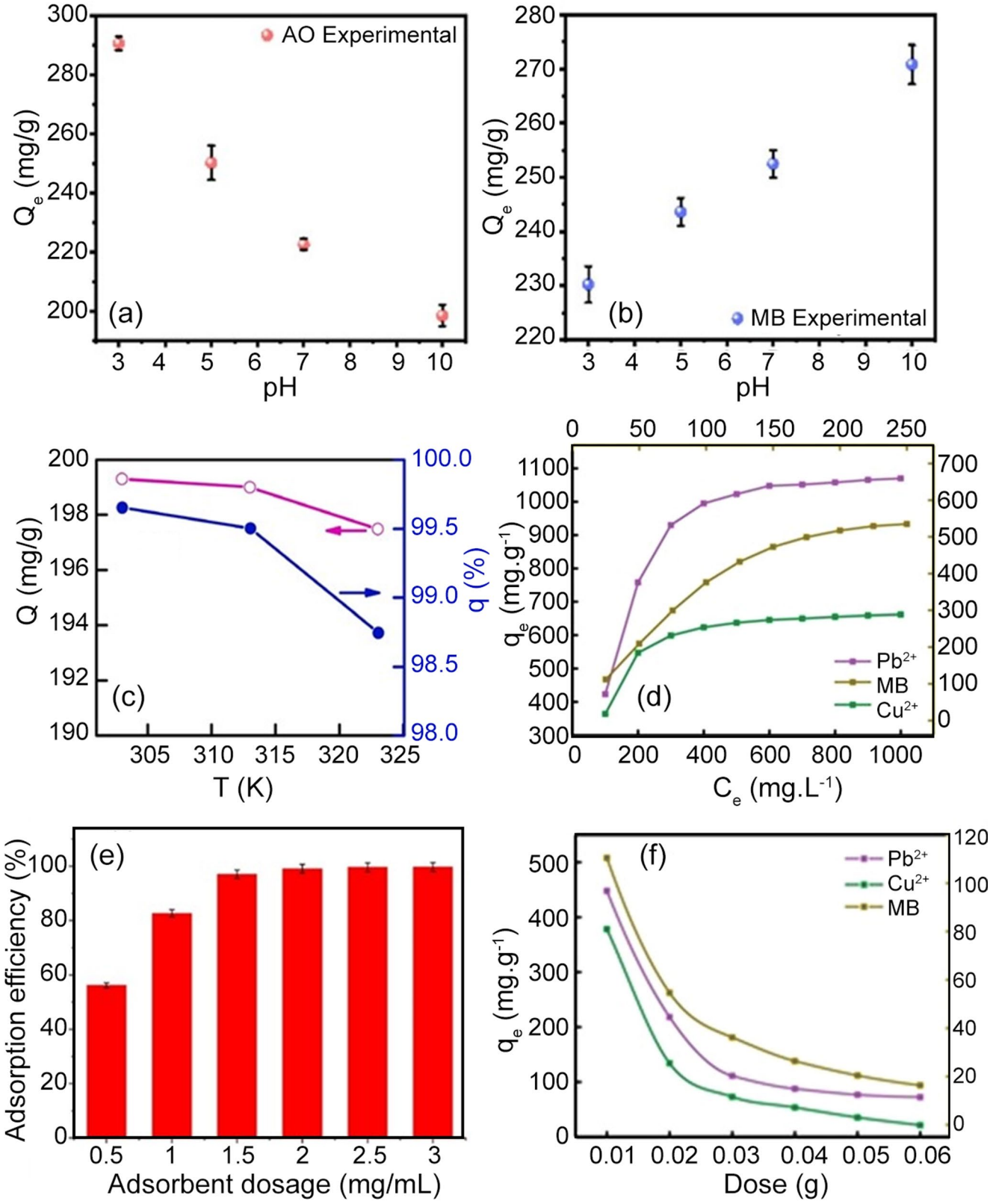



**Fig. 6** **a** AO and **b** MB adsorption capacity under various pH conditions (C0 = 300 mg/L; contact time = 6 h; temp = 25 °C; volume of dye solution = 30 mL; adsorbent dose = 30 mg; n = 3) [101], *Reproduced with permission from Elsevier,* **c** impact of temperature on CCH-TS adsorption for MB [109], *Reproduced with permission from Elsevier,* **d** effect of pollutant conc. using CS/EDTA/CBC adsorbent for MB adsorption [111], *Reproduced with permission from Elsevier*, **e** impact of ACCB Dosage on diclofenac adsorption [108], *Reproduced with permission from Elsevier*, and **f** influence of adsorbent amount for MB removal [111], *Reproduced with permission from Elsevier*

made from activated carbon and chitosan. Increasing the dosage typically enhances the removal of dyes and other organic pollutants attributed to availability of more active binding sites on the composite surface. This also provides useful insight into the cost-effectiveness of the process, as an optimal dosage can achieve high removal rates while minimizing material use [172–174]. Recently, multiple research workers have explored how varying the adsorbent dosage affects the removal efficiency of organic contaminants.

Kumar et al. investigated how varying the dosage of ACCB (activated carbon-chitosan beads) influenced the adsorption of diclofenac (DIF). The study was conducted using a 100 mg/L DIF solution at pH 6.0, with adsorbent dosages varies from 0.5 to 3.0 mg/mL, as shown in Fig. 6e. The findings revealed that an optimal 97.02% DIF removal occurred at a dose of 1.5 mg/mL. Additionally, the removal efficiency increases with higher ACCB dosage due to more available active sites but reaches equilibrium as the DIF molecules become limited for further adsorption [108].

Similarly, Fan et al. studied the impact of varying dosages on cotton charcoal/chitosan biomass-based hydrogel for MB adsorption (shown in Fig. 6f). Their findings reveal that increasing the adsorbent quantity leads to decline in uptake capacity for MB. This is because more adsorbent provides additional sites that remain unused due to a constant pollutant concentration, thus lowering overall capacity. Nonetheless, the removal rate of pollutants rises with greater adsorbent amounts [111]. These results confirm that optimizing the adsorbent dosage is essential for achieving high efficiency and practical use of activated carbon/chitosan-based composites in treating wastewater.

### Ionic strength

The ionic strength is a key factor influencing the adsorption behavior of biomass-derived activated carbon/chitosan hydrogel composites. Variations in ionic strength can influence the electrostatic interactions between the composite surface and the pollutants, thereby either promoting or reducing adsorption. The impact of ionic strength largely depends on nature of pollutant and functional groups present on the adsorbent. Recently, several researchers have investigated the effect of ionic strength on adsorption performance of organic and inorganic pollutant. For instance, Al-Hazmi et al. investigated how ionic strength influences the elimination of Ni(II) ions using activated algae embedded within a chitosan/carboxymethyl cellulose framework hydrogel beads. To assess this, NaCl was introduced to the Ni (II) solution at concentration ranging from 10 to 40 g/L, the uptake capacity decreased as ionic strength increased, with a significant decline observed at 40 g/L $Na^+$. This demonstrated that higher concentrations of $Na^+$ ions interfere with the electrostatic attraction between Ni(II) ions and the prepared hydrogel beads These results provide further evidence of underlying chemical interactions between the hydrogel matrix and the metal ions [153].

In summary, adsorption performance of biomass-derived activated carbon/chitosan hydrogel composites is governed by multiple operational parameters. pH influences surface charge and functional group reactivity; temperature affects adsorption kinetics and thermodynamics; pollutant concentration dictates mass transfer rates; adsorbent dosage determines the number of active sites; and ionic strength modulates electrostatic interactions. Optimal tuning of these factors is crucial to enhance the efficiency of contaminant removal and ensuring practical application in wastewater treatment.

## Adsorption of inorganic pollutants (heavy metals)

Heavy metals are elements known for their high atomic weights and densities, encompassing elements such as arsenic (As), chromium (Cr), cadmium (Cd), mercury (Hg), and lead (Pb). These substances present significant threats to both the environment and human health due to their toxic nature and ability to persist and accumulate in ecosystems and biological tissues [175, 176]. The toxicity of these metals differs depending on the type of metal, its concentration, duration of exposure, and the pathway through which it enters the body. Long-term exposure to elevated levels of heavy metals can lead to severe health issues, including damage to the brain, kidneys, liver, and respiratory system [176]. Heavy metal contamination poses a serious threat to ecosystems. These pollutants can accumulate in soils, water, and sediments, leading to ecological imbalances and biodiversity loss. A wide range of methods have been developed and applied to eliminate heavy metal ions from various environmental sources. These approaches effectively lower heavy metal concentrations to permissible levels, thereby contributing to the protection of both ecological systems and public health [176]. Adsorption is the most commonly used method because it is easy to operate and inexpensive [177]. However, biomass and natural inorganic materials, such as clay minerals, metal phosphates, zeolites, modified saps, kaolin, fly ash, and biosorbents like agricultural waste, peat, and microorganisms, have been employed as adsorbents. Among them, activated carbon stands out as one of the most widely used materials for heavy metal elimination. Its surface functional groups are crucial in influencing its ability to bind with metal ions [178, 179]. AC are commonly used in water treatment to remove heavy metals owing to highly porous architecture and high surface area, and diverse chemical properties. The presence of various surface functional groups, including phenolic, carbonyl,

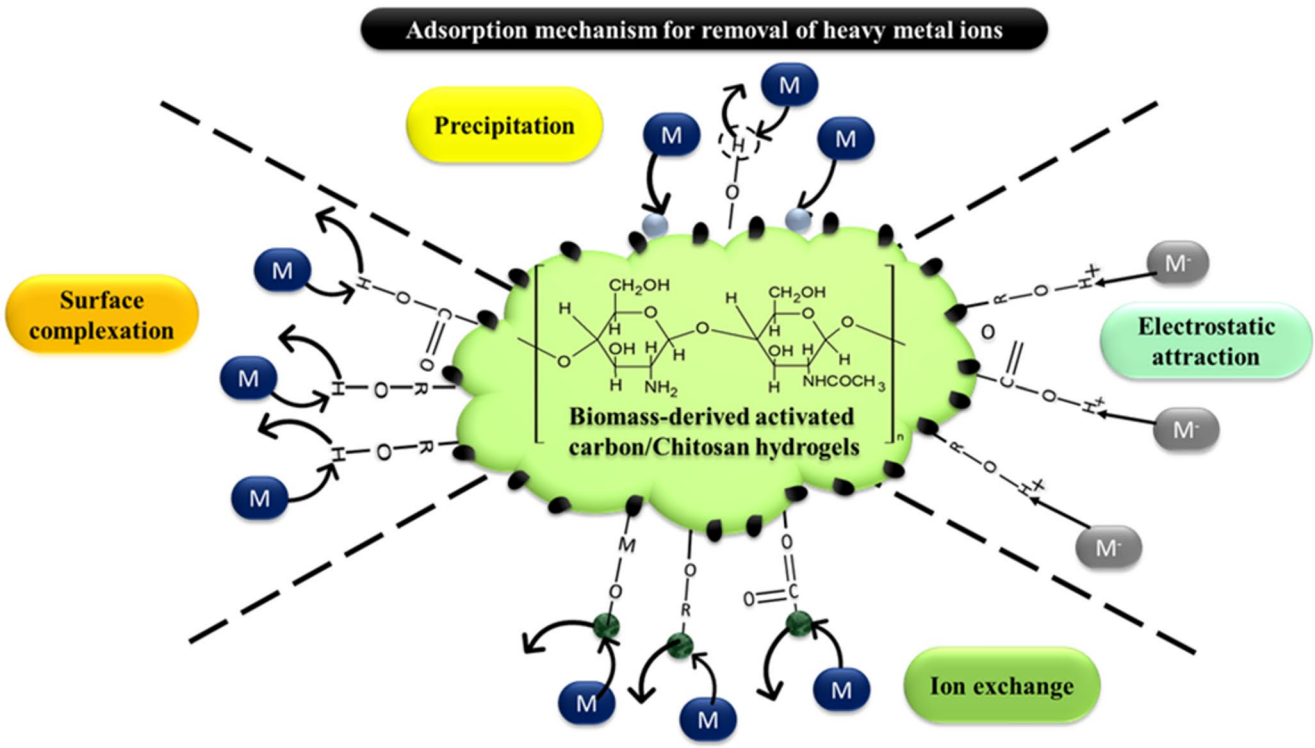



**Fig. 7** Adsorption mechanism for toxic metal ions removal

lactonic, carboxylic, and quinonoid moieties, enhances their ability to effectively bind and capture metal ions during the adsorption process [180]. Activated carbon also has some drawbacks, like being expensive and potentially causing secondary pollution. Consequently, chitosan (CS) has gained attention as a potential substitute adsorbent for treating wastewater [181]. It is a great biomaterial for removing metal ions from water, attributed to its non-toxic nature, biodegradable, and has useful functional groups like amino and hydroxyl groups. These groups act as active sites that help capture and eliminate heavy metal ions from the water [158, 182]. Chitosan eliminates heavy metals primarily through two distinct mechanisms: physical and chemical adsorption. The physical adsorption involves interactions such as electrostatic attraction and hydrogen bonding, which are relatively easy to control through external conditions, enabling efficient separation of heavy metals from the adsorbent. In contrast, chemical absorption involves stronger coordination bonds and is irreversible, making it difficult to recover and reuse the adsorbent. The 3D porous structure of hydrogels helps heavy metals diffuse into the material, improving adsorption efficiency. Since CS has a strong affinity for metal ions, these hydrogels effectively adsorb $Cu^{2+}$, $Fe^{2+}$, and $Pb^{2+}$ [158]. An illustration of the adsorption process is presented in Fig. 7, which illustrates the key adsorption mechanisms electrostatic attraction, surface complexation, ion exchange, and precipitation, through which biomass-derived activated carbon/chitosan hydrogels effectively remove heavy metal ions. The effectiveness of metal ion removal is largely attributed to the presence of reactive functional groups on chitosan and the highly porous architecture of activated carbon, both of which promote strong interactions and facilitate the adsorption of contaminants from water. However, combining activated carbon with chitosan hydrogel creates a highly efficient hybrid material that leverages the strengths of both components. This combination not only enhances the overall adsorption capacity for heavy metals but also capitalizes on the favorable characteristics of each material, leading to improved performance in water treatment applications. In recent studies, researchers have explored the potential of hydrogels composed of activated carbon and chitosan as promising materials for the removal of hazardous metal ions from contaminated water sources, and some comparative analyses using biomass-derived activated carbon/chitosan hydrogels for removing heavy metals through the adsorption technique are described below and summarized in Table 2 above, highlighting their synthesis methods, targeted pollutants, adsorption capacities, mechanisms and kinetic models.

For example, Dao and colleagues investigated the performance of magnetic activated carbon/chitosan hydrogel beads for the removal of Ni (II) ions. Their findings indicated that, under optimal conditions specifically at 25 °C, pH 6, an adsorbent dosage of 1.0 g/L, and a contact duration

of 6 h, the material achieved a peak uptake capacity of 108.70 mg/g. Additionally, reusability assessments indicated that the beads retained over 86.25% of their removal efficiency even after five cycles of use. The underlying adsorption process was due to several interactions, like ion exchange, complex formation, and electrostatic attractions [147].

In another study, Fan et al. prepared a hydrogel composite using cotton-derived charcoal and chitosan biomass to eliminate $Pb^{2+}$ and $Cu^{2+}$ ions from water. The results demonstrated remarkable adsorption capacities, reaching 1105.78 mg/g for $Pb^{2+}$ and 678.04 mg/g for $Cu^{2+}$. These high values were attributed to strong chemical interactions between the hydrogel and metal ions, resulting stable chelating structures. The re-usability tests were also evaluated for both ions indicating that the prepared hydrogel-maintained adsorption effectiveness of 80% for lead and 79% for copper after five cycles. Additionally, the recovery rate for both metal ions remained above 89% after these cycles [111].

Puspitasari et al. investigated the removal of heavy metals like Cu, Zn, Fe, and Pb utilizing chitosan hydrogel-based activated carbon nanoparticles. The results demonstrated the effective removal efficiency of 32.65% for absorbing lead [183]. Similarly, Mirzayant et al. studied the removal of iron (Fe) ions utilizing activated carbon coated with chitosan gel. The synthesized material showed an impressive removal capacity of 2.076412 mg/g. The adsorption isotherm analysis revealed high correlation coefficient ($R^2$) of 0.9956, indicating excellent fit with the Langmuir model, effectively representing the adsorption behavior [149].

Sharififard et al. utilized activated carbon chitosan hydrogel beads for removing platinum (Pt) and palladium (Pd) ions. The results demonstrated that the material achieved peak adsorption capacities of 52.63 mg/g for platinum and 43.48 mg/g for palladium. The adsorption data closely matched the Langmuir isotherm model, while, the kinetic evaluation revealed that the adsorption process adhered to a pseudo-second-order model, suggesting chemisorption as the rate-limiting step [148].

Al Hazmi et al. explored the adsorption performance of hydrogel beads synthesized from chitosan and carboxymethyl cellulose (CMC), embedded with activated algae, for the removal of Ni(II) ions from aqueous media. Kinetic analysis confirmed that the adsorption process conformed to a pseudo-second-order model, implying that chemical interactions between the metal ions and the adsorbent played a dominant role in the rate-limiting step. Equilibrium data exhibited a strong fit with the Langmuir isotherm, indicating monolayer adsorption on a homogeneously distributed surface. Thermodynamic evaluation demonstrated that the process was both endothermic and spontaneous, as evidenced by the enhanced adsorption capacity at elevated temperatures. Optimal adsorption was achieved at a solution pH of 5, using 0.02 g of prepared hydrogel beads in 25 mL of Ni(II) solution, resulting in a maximum adsorption capacity of 301.05 mg/g [153].

Hajri et al. engineered a chitosan-based hydrogel composite incorporating activated algae and a metal–organic framework (MOF) to facilitate the adsorption of Co(II) ions from aqueous solutions. The material demonstrated a notably high adsorption capacity, reaching 445.8 mg/g. Kinetic studies indicated that the adsorption process adhered to a pseudo-second-order model, suggesting chemisorption as the dominant mechanism. Furthermore, equilibrium data aligned closely with the Langmuir isotherm, reflecting monolayer adsorption behavior with an associated adsorption energy of 31.8 kJ/mol. The process was both endothermic and spontaneous, as higher temperatures enhanced metal uptake. Reusability tests over five adsorption–desorption cycles revealed a gradual decline in performance, with adsorption capacities of 98.2%, 96.8%, 92.6%, 89.4%, and 87.6% of the initial value, respectively. This reduction was attributed to structural changes in the hydrogel and a possible loss of active adsorption sites [154].

Overall, biomass-derived activated carbon/chitosan hydrogel composites exhibit excellent performance in adsorbing various heavy metal ions due to the synergistic effects of chitosan's functional groups and activated carbon's high surface area. The combination of electrostatic attraction, ion exchange, and surface complexation mechanisms enables selective and efficient removal of toxic metals. Additionally, many of these adsorbents demonstrate good regeneration and reusability potential, making them promising candidates for sustainable water purification applications.

## Synergistic and competitive adsorption in complex wastewater systems

In real-world wastewater treatment scenarios, pollutants rarely exist in isolation. Instead, wastewater often contains complex mixtures of organic and inorganic contaminants, like colorants, pharmaceuticals, pesticides, and metal ions. The simultaneous existence of these pollutants can significantly impact the adsorption performance of composite materials due to competitive or synergistic interactions at the adsorption sites. While many studies, including the current review, focus on single-pollutant systems (e.g., methylene blue or PNP), real wastewater often contains multiple co-existing pollutants. These pollutants may interact competitively or synergistically during the adsorption process, influencing the overall removal efficiency. Several studies have shown that biomass-derived activated carbon/chitosan hydrogel composites can effectively adsorb mixed

contaminants like dyes, metal ions or antibiotics, though the efficiency depends on factors like pollutant concentration, surface functional groups, and solution pH.

For example, Zaoui et al. investigated study on the simultaneous removal of methylene blue (MB) and orange G (OG) dyes using chitosan/activated carbon/metal oxide composite hydrogel. The results demonstrated significantly high uptake potential of 323.625 and 134.589 mg/g for OG and MB, emphasizing excellent performance of the composite material. The remarkable adsorption efficiency was attributed to the synergistic incorporation of activated carbon, metal oxides, and the chitosan biopolymer. Compared to pristine chitosan beads, the composite beads exhibited superior adsorption attributed to the enhanced number of active sites, increased surface area, and improved porosity provided by the embedded components. The incorporation of activated carbon contributed to enhanced π–π stacking interactions, while the metal oxides introduced ion exchange capabilities, further expanding the spectrum of adsorption mechanisms. Chitosan served as a biopolymeric matrix that not only facilitated the dispersion of the active components but also enabled easy separation and potential reuse of the beads. The study highlighted that among the different interactions involved, electrostatic attraction was the most dominant. The cationic MB dye interacted preferentially with negatively charged groups like $-NH^-$, $-COO^-$, and $-O^-$ on the adsorbent surface, while the anionic OG dye exhibited strong affinity toward positively charged sites, particularly the protonated $-NH_3^+$ groups and metal cations like $Ca^{2+}$ and $Al^{3+}$. π–π interactions also contributed significantly, especially due to the aromatic domains of activated carbon, which aligned with the conjugated systems of both dyes. Furthermore, ion exchange mechanisms played a moderate but crucial role, where in metal ions such as $Ca^{2+}$ and $Al^{3+}$ were exchanged with dye ions in the solution. Overall, the synergistic combination of multiple functional materials significantly improved the adsorption behavior of the composite beads, making them promising candidates for treating dye-contaminated wastewater [100].

Fan et al. explored the sorption behavior of $Pb^{2+}$, $Cu^{2+}$, and methylene blue (MB) using a functionalized cotton-based charcoal/chitosan hydrogel composite (CS/EDTA/CBC). The composite material demonstrated outstanding adsorption capacities, indicating 1105.78 mg/g for $Pb^{2+}$, 678.04 mg/g for $Cu^{2+}$, and 590.72 mg/g for methylene blue (MB), highlighting its effectiveness in multi-pollutant remediation. To assess selectivity and competitive behavior of the adsorbent, binary and ternary adsorption experiments were conducted. The results showed mutual inhibition among the three pollutants during co-adsorption, indicating competition for active sites. However, the adsorbent exhibited preferential uptake, with $Pb^{2+}$ showing the highest competitive adsorption, followed by $Cu^{2+}$, while MB displayed the lowest affinity. This trend correlates with the electronegativity of the species, as higher electronegativity facilitates stronger attraction to the adsorbent surface. $Pb^{2+}$, being the most electronegative among the three, interacted more readily with active sites compared to $Cu^{2+}$ and MB [111].

## Challenges and future directions

Biomass-derived activated carbon and chitosan hydrogels have emerged as promising materials for removing both organic and inorganic pollutants from wastewater owing to its sustainability, cost-effectiveness, and multi-functionality. However, several challenges hinder their full potential in large-scale environmental applications. Overcoming these challenges is essential to improving their performance and making them practical solutions for water purification and pollutant removal.

A major challenge is the inconsistency in the characteristics of activated carbon because of the diverse nature of biomass feedstocks such as agricultural residues, forestry by-products, and algal biomass. These differences result in variations in pore structure, surface area, and functional groups, which directly influence adsorption capacity and selectivity. Standardized pre-processing steps like alkaline treatment, controlled temperature pyrolysis, and chemical activation using KOH or $ZnCl_2$ can help reduce variability and produce more reliable adsorbents. Future research should focus on creating clear guidelines for choosing the most suitable types of biomasses, depending on the specific pollutants targeted for removal.

Another key challenge lies in the structural limitations of chitosan hydrogels. While they contain plenty of functional groups ($-NH_2$ and -OH), which offer good affinity toward multiple pollutants, their mechanical strength and chemical stability often fall short especially under varying pH and salinity conditions. To make them durable for long term use by enhancing hydrogel strength through chemical crosslinking (e.g., using agents such as glutaraldehyde) or integration with nanofillers such as graphene oxide, silica nanoparticles, or clay can significantly boost the hydrogel strength and stability without sacrificing adsorption performance.

The regeneration and reusability of biomass-derived AC and chitosan hydrogels also remains a crucial area for development. Although many studies highlighted their impressive initial adsorption capacities, maintain this performance over multiple use cycles is a challenge. Sustainable regeneration methods such as low-temperature thermal desorption, alkali-acid washing, or microwave-assisted regeneration, need to be explored and optimized for maintaining performance over multiple cycles. Studies should also quantify

adsorption efficiency after each regeneration cycle to assess long-term effectiveness and practical value in real world applications.

Another important aspect is the environmental fate of used adsorbents. Although these materials are derived from renewable sources, their end-of-life handling must be environmentally responsible. Life cycle assessments (LCA) and ecotoxicity evaluations should be conducted to understand potential secondary pollution, especially in cases where nanocomposites or metal oxides are used.

In practical applications, wastewater is rarely composed of a single pollutant. Instead, it contains a complex mixture of organic dyes, heavy metals, pharmaceuticals, and other substances that can interfere with one another during adsorption. Competitive adsorption in such multicomponent systems often reduces overall efficiency. Therefore, future studies must examine the adsorption behavior in binary and ternary systems, simulate real wastewater conditions, and design materials with enhanced selectivity. Tailoring the surface chemistry of composites using functionalized adsorbents may improve affinity toward specific contaminants even in complex matrices.

Furthermore, the scale-up of production processes remains a bottleneck. Many synthesis routes are energy-intensive or involve hazardous chemicals. Adopting green synthesis methods (e.g., hydrothermal carbonization or biotemplating) and continuous-flow production systems can support industrial-scale fabrication without compromising material quality.

While laboratory studies demonstrate the high potential of biomass-derived activated carbon and chitosan hydrogels, their application in real-world scenarios is often constrained by practical limitations. These include high production and processing costs, challenges in large-scale material handling, and the lack of standardized protocols for performance assessment. Moreover, the environmental risks associated with residual composite materials, such as nanoparticle leaching or secondary contamination, must be carefully evaluated through ecotoxicological studies. Regulatory approval for using these materials in water purification also remains a challenge. Concern about their safety, long-term stability, and disposal protocols must be addressed to meet governmental and environmental standards. Future studies should include through cost–benefit analyses, detailed risk assessments, and active engagement with regulatory guidelines to ensure that these materials are not only effective in lab but also safe, economically viable, and practical for widespread use in real-world wastewater treatment systems.

In summary, while biomass-derived activated carbon and chitosan hydrogels show great promise for removing both organic and inorganic pollutants, there are some challenges to address in order to fully harness their potential. Future research should aim to improve physical properties, boosting mechanical strength, refining regeneration techniques, and developing cost effective methods for large scale production. Exploring hybrid materials and advanced functionalization approaches could open up new opportunities for targeted pollutant removal and environmental remediation. With continued innovation and strong collaboration between researchers, industries, and policymakers, these materials could play a key role in advancing the future of sustainable and eco-friendly water treatment technologies.

## Conclusion

Biomass derived activated carbon chitosan hydrogel composites have gained significant attention as sustainable and efficient material for environmental remediation. These composites effectively remove a broad range of organic and inorganic pollutants from water. The combination between activated carbon, known for its high surface area and strong adsorption ability and chitosan, a natural biodegradable polymer with excellent capablility of forming hydrogels results in composites with improved adsorption efficiency, structural stability, and ecofriendliness. Among the various adsorbents reviewed, some of these composites exhibit remarkably excellent adsorption capacities, reaching up to 1428 mg/g for Food Blue 2 and 1370 mg/g for Food Red 17. These high performances are mainly attributed to mechanisms like electrostatic interactions, π–π stacking, and pore filling effects. Similarly, cotton-based charcoal/chitosan hydrogels showed notable uptake of heavy metals, with $Pb^{2+}$ reaching 1105.78 mg/g, largely due to surface complexation and electrostatic attraction. The choice of fabrication method plays a critical role in tailoring the structural and functional properties of these composites. Techniques such as ultrasonication and free radical polymerization have been particularly effective in creating porous, high-performance materials. Conversely, some adsorbents with limited surface functionality or pore structure, like date palm seed-derived activated carbon/chitosan hydrogel microbeads, chitosan-carbon hydrogel beads and chitosan-activated carbon/zinc oxide hydrogel beads showed lower adsorption capacities, highlighting the need for thoughtful material design. Most composites adhere Langmuir isotherm and pseudo-second-order kinetic models, indicating monolayer chemisorption as the dominant mechanism. Additional interactions, including hydrogen bonding, acid–base reactions, and intraparticle diffusion, also contributed to the overall adsorption process. While these materials offer several advantages, including renewability, cost-effectiveness, and multifunctionality challenges remain in achieving consistent material

quality, integrating diverse components, and ensuring recyclability and regeneration. Addressing these limitations will require further research into standardized fabrication protocols, optimization of functional additives, and exploration of novel biomass sources. Finally, biomass-derived activated carbon/chitosan hydrogels hold great potential for scalable and eco-friendly wastewater treatment. Continued innovations in their design and use could significantly contribute to the growth of efficient, cost effective, and green solutions for tackling water pollution and achieving broader environmental sustainability goals.

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**Data availability** No new data was used for the research described in the article.

## Declarations

**Conflict of interests** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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